Introduction to Methods Engineering and Operations Analysis

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The techniques discussed in this chapter have the following basic objectives:

- To analyze existing work systems
- To make improvements in existing work systems
- To design new work systems
- To plan the facilities in which the work systems operate

Most of the techniques that we discuss are traditional industrial engineering "tools of the trade." Methods engineering is often associated with work measurement. The two areas are often referred to collectively as "motion and time study." Many of the motion and time study techniques can be traced to the origins of industrial engineering, when it was referred to as the "scientific management movement." The techniques have been used to analyze, design, and measure work for many decades.

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Methods engineering is a broader term than motion study. The scope of this field has expanded well beyond its original focus on human body motions to perform physical labor. We define ***methods engineering*** as the analysis and design of work methods and systems, including the tooling, equipment, technologies, workplace layout, plant layout, and environment used in these methods and systems.[1] Other names have sometimes been used to indicate the same basic approaches of methods engineering. These other names include ***work study***, ***work simplification***, ***methods study***, ***process re-engineering***, and ***business process re-engineering***.

The traditional objectives in methods engineering, whatever the name for this improvement activity, are the following:

- To increase productivity and efficiency
- To reduce cycle time
- To reduce product cost
- To reduce labor content

In addition, other objectives are frequently defined for process improvement efforts. Some of these additional objectives have a more contemporary appeal in today's society. They include the following:

- To improve customer satisfaction
- To improve product and/or service quality
- To reduce lead times and improve work flow
- To increase work system flexibility
- To improve worker safety
- To apply more ergonomic work methods
- To enhance the environment (both inside and outside the facility)

A term closely related to methods engineering is ***operations analysis***, defined as the study of an operation or group of related operations for the purpose of analyzing their efficiency and effectiveness so that improvements can be developed relative to specified objectives.[2] The specified objectives are basically the same as in methods engineering: to increase productivity, reduce time and cost, and improve safety and quality. Thus, methods engineering and operations analysis are very similar terms, except that methods engineering places more emphasis on design. Both terms are widely used in industrial engineering and this is why both names are included in our chapter title.

[1]The definition of methods engineering developed by the Work Measurement and Methods Standards Subcommittee [ANSI Standard Z94.11-1989] is the following: "That aspect of industrial engineering concerned with the analysis and design of work methods and systems, including the technological selection of operations or processes, specification of equipment type and location, design of manual and worker-machine tasks. May include the design of controls to insure proper level of output, inventory, quality, and cost."

[2]The definition of operations analysis developed by the Work Measurement and Methods Standards Subcommittee [ANSI Standard Z94.11-1989] is the following: "A study of an operation or scenes of operations involving people, equipment, and processes for the purpose of investigating the effectiveness of specific operations or groups so that improvements can be developed which will raise productivity, reduce costs, improve quality, reduce accident hazards, and attain other desired objectives."

1 EVOLUTION AND SCOPE OF METHODS ENGINEERING

The initial research in the area of methods engineering by Frank Gilbreth in 1885 dealt with the motions performed by workers in bricklaying. Motion study was truly an appropriate term for Gilbreth's research. The study of manual physical labor in manufacturing and construction was the primary concern of the scientific management movement in the late nineteenth and early twentieth centuries, and motion study was one of the two principal techniques used by the practitioners in that movement (the other, of course, was time study). Today, methods engineering is being applied in many other areas of work, including indirect labor, logistics, service operations, office work, and plant layout design. As these other areas have grown in importance in the economies of industrialized nations, methods engineering has been applied to analyze, improve, and design the work methods.

In terms of the problems that are addressed, methods engineering can be divided into two areas: (1) methods analysis and (2) methods design. ***Methods analysis*** is concerned with the study of an existing method or process, usually by breaking it down into the work elements or basic operations that comprise it.[3] By examining the details of the elements or operations, a systematic search can be carried out to find ways to improve the method or process. The systematic search often consists of checklists of questions and suggestions that offer opportunities for improvement. There are several objectives of using checklists in methods analysis:

- To *eliminate* unnecessary and non-value-adding elements or operations from the larger method or process
- To *combine* multiple elements or operations by performing them at one location and/or simultaneously
- To *rearrange* the elements or operations into a more logical sequence and work flow
- To *simplify* the remaining elements or operations so they can be accomplished more quickly and with minimum effort.

In addition to studying an existing method or process, methods analysis can also be used to analyze a proposed new method for possible improvements. In this regard the two areas of methods analysis and methods design overlap each other.

Methods design is concerned with either of the following situations: (1) the design of a new method or process or (2) the redesign of an existing method or process based on a preceding methods analysis. The design of a new method or process occurs when a new product or service is introduced and/or a new facility or equipment is installed, and there is no existing precedent for the operation. In this case, the method or process must be designed from scratch. This can often be accomplished by referring to best current practice for similar operations and attempting to improve on the current work design. In other cases, an original work design must be developed, including the basic operations,

[3]The definition of methods analysis developed by the Work Measurement and Methods Standards Subcommittee [ANSI Standard Z94.11-1989] is the following: "That part of methods engineering normally involving an examination and analysis of an operation or a work cycle broken down into its constituent parts for the purpose of improvement, elimination of unnecessary steps, and/or establishing and recording in detail a proposed method of performance."

methods, equipment, special tooling, workplace layout, and plant layout. As in other areas of design, guidelines are available, such as the principles of motion economy and the plant layout planning approach, and these can be applied in solving the design problem.

Methods engineering studies have traditionally been the province and responsibility of industrial engineers. Today, methods analysis and design are no longer accomplished exclusively by industrial engineers. They are accomplished, with greater or lesser effectiveness, by a variety of individuals, departments, and teams. The workers themselves often participate in the development and improvement of their own work methods, under the assumption that they know their jobs better than anyone else. Industrial engineers often participate in these improvement activities, serving an important role as technical consultant and mentor.

2 HOW TO APPLY METHODS ENGINEERING

In this section we present a systematic approach that can be used to accomplish a methods engineering study. We then survey the tools and techniques used in such a study. Finally, we present a procedure for selecting among alternative possible solutions to a methods engineering problem.

2.1 Systematic Approach in Methods Engineering

An underlying assumption in methods engineering is that a systematic approach is more likely to yield operational improvements than an undisciplined approach. Our systematic approach to problem solving in methods engineering has its basis in the ***scientific method*** used in science, research and development, engineering design, and other problem areas. The systematic approach in methods engineering consists of the steps described below.

Step 1: Define the Problem and Objectives. The problem is the reason for needing a systematic approach to determine its solution. The problem in a methods engineering study may be low productivity, high cost, inefficient methods, or the need for a new method or a new operation. The objective is the desired improvement or new methods design that would result from the methods engineering project. Possible objectives are to increase productivity, reduce labor content and cost, improve safety, or develop a new method or new operation. These are the typical objectives discussed in our chapter introduction. The problem definition and objectives must be specific to the problem under investigation, although there may be similarities with other problems.

Step 2: Analyze the Problem. This step consists of data collection and analysis activities that are most appropriate for the type of problem being studied. The kinds of activities often used in this step include the following:

- Identify the basic function of the operation.
- Gather background information.
- Observe the existing process or observe similar processes if the problem involves a new work design.

- Collect data on the existing operation and document the details in a format that lends itself to examination.
- Conduct experiments on the process.
- Develop a mathematical model of the process or utilize an existing mathematical model such as those on work systems and how they work.
- Perform a computer simulation of the process.
- Use charting techniques.

A survey of the analysis techniques used in step 2 is provided in Section 2.2.

Step 3: Formulate Alternatives. There are always multiple ways to perform a task or accomplish a process, some of which are more efficient and effective than others. Only by enumerating the alternative ways and comparing them can the most efficient and effective method or process be determined.[4] However, the purpose of this step in the problem-solving approach is not to identify the best alternative but to formulate all of the alternatives that are feasible.[5]

Step 4: Evaluate Alternatives and Select the Best. This step consists of a methodical assessment of the alternatives and the selection of the best solution among them, based on the original definition of the problem and objectives. The selection procedure described in Section 2.3 is useful in this step of the methods engineering approach.

Step 5: Implement the Best Method. Implementation means installing the selected solution: introducing the changes proposed in the existing method or operation, or instituting the new method or process. This may involve pilot studies or trials of the new or revised method preliminary to online implementation and application of the method. Implementation also includes complete documentation of the new or revised method and replacement of the previous documentation in the case of a revised method. Unless the old documentation is replaced, the old method may remain as the official method until a new methods engineering study is performed sometime in the future (thus reinventing the wheel).

Step 6: Audit the Study. It is desirable to perform an audit or follow-up on the methods engineering project. How successful was the project in terms of the original problem definition and objectives? What were the implementation issues? What should be done differently in the next methods engineering study? For an organization committed to continuous improvement, answers to these kinds of questions help to fine-tune its problem-solving and decision-making skills.

[4]In the scientific method as applied to research, the alternatives are typically the different proposed theories to explain the observed phenomenon. Ultimately, one theory is selected as the most reasonable in terms of explaining the phenomenon.

[5]Even the infeasible solutions should probably be included in the list. Sometimes the alternatives that initially seem infeasible turn out to be quite doable after all. What made them seem unacceptable on first consideration were artificial or false constraints that proved not to be constraints upon closer scrutiny.

2.2 The Techniques of Methods Engineering

A variety of techniques are available for operations analysis. The techniques are most closely associated with the analysis step in methods engineering, although they may also be applicable in some of the other steps as well. As a starting point, we have the basic data collection and analysis techniques discussed in Section 3. These are graphical and statistical methods for gathering, plotting, and displaying data. They include histograms and *x-y* plots as well as other charts and diagrams.

Beyond the basic data collection and analysis techniques, which are applicable in many disciplines other than methods engineering, there are the specialized analysis techniques more closely associated with operations analysis and industrial engineering. They are described briefly in the following sections.

Charting and Diagramming Techniques. There are many charting techniques available for collecting, displaying, and analyzing data on a given work system or operation sequence of interest. They can be classified into the following categories:

- *Network diagrams.* These are used for analyzing work flow, assembly line balancing, and project scheduling. Special algorithms are often available to analyze these network diagrams.
- *Traditional industrial engineering charting techniques.* These are used to symbolize and summarize the details of an existing operation or sequence of operations. The traditional charting techniques can be used to analyze the activities of one human worker, groups of workers, worker-machine systems, materials, parts, and products.
- *Block diagrams and process maps.* These diagrams represent alternative ways of depicting processes. They are sometimes used in place of the traditional IE charting techniques.

Motion Study and Work Design. This area of methods engineering is concerned with the study of the basic motions of a human worker while performing a given task. The basic motions include ***reach*** (using the hand to reach for an object), ***grasp*** (grasping the object), ***move*** (moving the object), and ***release*** (releasing it). All manual tasks performed at a single workplace are composed of these basic motions. There are 17 basic motion elements, most of which involve movements of the arm and hand. By studying the basic motions in the work method used by a worker, unnecessary motions can be eliminated, or some of the motion elements can be combined (for example, using both hands to simultaneously perform motions rather than one arm doing everything), or the method can be otherwise simplified.

Over many years, the study of basic motions in manual operations has resulted in the development of certain principles on how to perform work. Commonly called the ***principles of motion economy***, they provide guidelines for work design in three categories: (1) use of the human body in developing the standard method, (2) workplace layout, and (3) design of the tooling and equipment used in the task. Many of the principles are

simple and obvious; for example, "design the work so that both hands are fully utilized." Yet simple work design principles such as these are often neglected in many manual operations performed throughout the world. By having these guidelines available, work methods can be designed to be safer, faster, more efficient, and less fatiguing.

Facility Layout Planning. A facility (e.g., factory, office building, warehouse, hospital) is a fixed asset of the organization that owns (or rents) it. ***Facility layout*** refers to the size and shape of a facility, the arrangement of the different functions and/or departments in it, and the way the equipment is positioned. The layout plays a significant role in determining the overall efficiency of the operations accomplished in the facility. Facility layout planning represents an important problem area in industrial engineering, and we discuss the techniques for solving it within the scope of methods engineering. The problem area includes designing a new facility, installing new equipment, retiring old equipment, and expanding (or contracting) an existing facility.

Plant layout planning and design are best accomplished using a systematic approach, similar to the systematic approach in methods engineering. As in other problem areas in methods engineering, there are tools available to use in a plant layout design project. The approach is called "systematic layout planning," developed by Richard Muther [13].

Work Measurement Techniques. Several of the work measurement techniques can also be used in methods engineering. We mention two that seem most relevant for studying and analyzing operations and methods: predetermined motion time systems and work sampling.

A ***predetermined motion time system*** (PMTS) is a database of basic motion elements and their associated normal time values, and it includes procedures for applying the database to analyze manual tasks and establish standard times for the tasks. The principal application of a PMTS is to determine standard times. However, some systems also include tools for analyzing methods and motions in the task, which is a methods engineering function.

Work sampling is a statistical technique for determining the proportions of time spent by workers or machines in various categories of activity. It can be applied to determine machine utilization, worker utilization, and the average time spent performing various types of activities. As such it can be a useful tool in methods engineering for identifying areas that need attention. For example, if a work sampling study finds that workers in a facility spend large amounts of their time waiting for work, then this is a management problem that should be addressed.

New Approaches in Methods Engineering. Nearly all of the preceding techniques have been used for many decades, some even longer.[6] More recently, alternative approaches for improving production and service operations have evolved from these traditional techniques. These alternative approaches include lean production (based on

[6]An example: Frank Gilbreth's interest in motion study dates from 1885; his book, *Motion Study*, was published in 1911.

the famed Toyota Production System), total quality management, and Six Sigma.

It is the view of this author that most of the tools used in these more recent approaches are basically adaptations and modifications of good industrial engineering techniques and principles. Not to demean their importance, they are often used with good effect to identify and solve problems and to make improvements. And they sometimes successfully serve the function of providing a rallying point for motivating workers who might not be motivated by something called motion and time study. Their objectives are generally the same as the objectives of methods engineering.

2.3 Selecting Among Alternative Improvement Proposals

Evaluating the alternative improvement proposals and selecting the best among them (step 4 in the systematic methods engineering approach) can be a difficult process in methods engineering. As always, it is helpful to use a systematic procedure to decide which improvement proposal(s) should be selected. The procedure recommended here is most applicable when the selection is to be made among specific proposals—for example, alternative process designs or different equipment proposals from vendors. The alternatives have been developed to address a specified problem or a need of the customer organization. Each proposed solution has its relative strengths and weaknesses, and the selection procedure must weigh the pros and cons in a fair and logical way. The procedure is explained as follows.[7]

Prior to the development of proposals, a list of technical features and functional specifications for the given application must be prepared by the customer organization. This is an essential part of the problem definition step (step 1) in the methods engineering approach. The operations analyst(s) or equipment vendors will use this list as the basis for formulating alternative solutions and proposals (step 3 in the systematic approach). The same list will be used by the customer organization to evaluate the proposals. The features and specifications should be divided into two categories:

1. *Must features.* These are the features and specifications that the proposal must satisfy. If these are not satisfied, then the proposal is not suitable for the application.
2. *Desirable features.* These are features and specifications that are not necessarily required to satisfy the application, but they are desirable.

After all of the proposals have been submitted, the evaluation process begins. The tabular format shown in Table 1, sometimes referred to as a ***criteria matrix***, is used. First, the proposals are evaluated against the must features. A candidate proposal must satisfy all of the must features or it is dropped from further consideration.

[7]Although we are presenting this selection procedure as a decision-making tool in methods engineering, its applications are much broader and more diverse. For example, it can be used for selecting which new car to buy, deciding which job offer to accept (assuming there is more than one offer), or even choosing a prospective spouse.

TABLE 1 Evaluation of Alternative Industrial Robots for a Welding Application in Example 1

	Industrial Robot Candidates			
	Model A	Model B	Model C	Model D
Must features:				
Continuous path control	OK	OK	OK	OK
Six-axis robot arm	OK	OK	Not OK	OK
Walkthrough programming	OK	OK	OK	OK
Desirable features:				
Ease of programming (0–9)	6	4		6
Capability to edit program (0–5)	4	2		5
Multipass features (0–4)	2	2		2
Work volume (0–9)	5	8		6
Repeatability (0–5)	5	2		4
Lowest price (0–5)	4	5		3
Delivery (0–3)	1	1		3
Evaluation of vendor (0–9)	6	5		8
Totals:	33	29		37

Source: [6].

Next, the proposals are compared against each of the desirable features. For each feature, a rating score is given to each candidate to indicate how well it satisfies that feature. No doubt there will be differences in the relative importance of the different features, and this is taken into account by assigning a maximum possible point score to each desirable feature. For example, if a particular feature is judged to be twice as important as another, then the more important feature is assigned a maximum score of, say, 10 points, while the less important feature is given a maximum of 5 points. These maximum point scores are judgment calls made by the methods engineer or by the collective wisdom of the project team doing the selection. Scoring each proposal on each feature is also a judgment call based on the relative merits of each candidate.

Finally, for the proposals that still remain after elimination of those that failed one or more of the must features, the scores of each proposal are tallied and the proposal with the highest total score is selected as the winner.

Of course, economics must play a role in the selection process. One might be inclined to simply choose the low cost bidder. However, that is often a mistake. Cost is rarely the only factor in implementing a new process, purchasing a piece of equipment, or making other investment decisions. The cost factor can be included as one of the desirable features in the list and given an appropriate weighting score to reflect its importance. The following example illustrates the use of this procedure to select a robot for a welding application.

Example 1 Selecting a Welding Robot[8]

Four industrial robots were being considered to satisfy an arc-welding application at the company. They are identified in Table 1 as Models A, B, C, and D. As suggested in the selection

[8]This example is based on an industrial case study in which the author participated. The case study was first reported in [7]. The company remains anonymous.

procedure, the features and specifications are divided into two categories: "must" and "desirable." The must features were considered essential for the application. The desirable features were assigned maximum point scores as shown in the table. The entries in the table for each robot indicate how that candidate was scored in each of the features. Note that one of the features was price, but it was not considered to be the most important feature.

Conclusion. First, model C was eliminated from consideration because it did not satisfy one of the "must" features. For the three remaining models, model D was selected because it had the highest point score among the desirable features. ■

3 BASIC DATA COLLECTION AND ANALYSIS TECHNIQUES

The data collection and analysis techniques discussed in this section are used by scientists, engineers, mathematicians, and statisticians. Most of them are statistical charting tools used to record and/or exhibit data so that it can be interpreted more readily. The techniques also have value for analysis. When used to measure and analyze production and service operations, these statistical tools are often associated with a field known as ***statistical process control*** (SPC).[9] Statistical process control and other quality-oriented programs are discussed in Chapter 21. In these programs, worker teams are organized to study operational problems. To enable the teams to be more effective, they are trained in the use of these basic tools. Thus, industrial engineers are not the exclusive users of these techniques in operations analysis. Instead, IEs must often serve as team leaders, training instructors, and expert consultants for the worker teams who carry out the studies.

Histograms. A ***histogram*** is a statistical graph consisting of bars representing different values or ranges of values, in which the length of each bar is proportional to the frequency or relative frequency of the value or range, as shown in Figure 1. Also known as a bar chart, it is a graphical display of the ***frequency distribution*** of the numerical data. What makes the histogram such a useful statistical tool is that it enables the analyst to quickly visualize the features of a complete set of data. These features include (1) the shape of the distribution, (2) any central tendency exhibited by the distribution, (3) approximations of the mean and mode of the distribution, and (4) the amount of scatter or spread in the data.

Example 2 Frequency Distribution and Histogram

Part dimension data from a manufacturing process are displayed in the frequency distribution of Table 2. The data are the dimensional values of individual parts taken from the process, while the process is running normally. Plot the data as a histogram and draw inferences from the graph.

Solution: The frequency distribution in Table 2 is displayed graphically in the histogram of Figure 1. We can see that the distribution is normal (in all likelihood), and that the mean is

[9]Although we are giving due recognition to the field of statistical process control, it should also be noted that many of these basic tools have been taught for many years in industrial engineering courses and used by practicing industrial engineers to study problems in methods engineering and operations analysis.

TABLE 2 Frequency Distribution of Part Dimension Data

Range of Dimension	Frequency	Relative Frequency	Cumulative Relative Frequency
$1.975 \leq x < 1.980$	1	0.01	0.01
$1.980 \leq x < 1.985$	3	0.03	0.04
$1.985 \leq x < 1.990$	5	0.05	0.09
$1.990 \leq x < 1.995$	13	0.13	0.22
$1.995 \leq x < 2.000$	29	0.29	0.51
$2.000 \leq x < 2.005$	27	0.27	0.78
$2.005 \leq x < 2.010$	15	0.15	0.93
$2.010 \leq x < 2.015$	4	0.04	0.97
$2.015 \leq x < 2.020$	2	0.02	0.99
$2.020 \leq x < 2.025$	1	0.01	1.00

around 2.00. We can approximate the standard deviation to be the range of the values (2.025–1.975) divided by 6, based on the fact that nearly the entire distribution (99.73%) is contained within $\pm 3\sigma$ of the mean value. This gives a σ value of around 0.008. ■

Pareto Charts. A ***Pareto chart*** is a special form of histogram, illustrated in Figure 2, in which attribute data are arranged according to some criteria such as cost or value. When appropriately used, it provides a graphical display of the tendency for a small proportion of a given population to be more valuable than the much larger majority. This tendency is sometimes referred to as ***Pareto's law***, which can be succinctly stated:

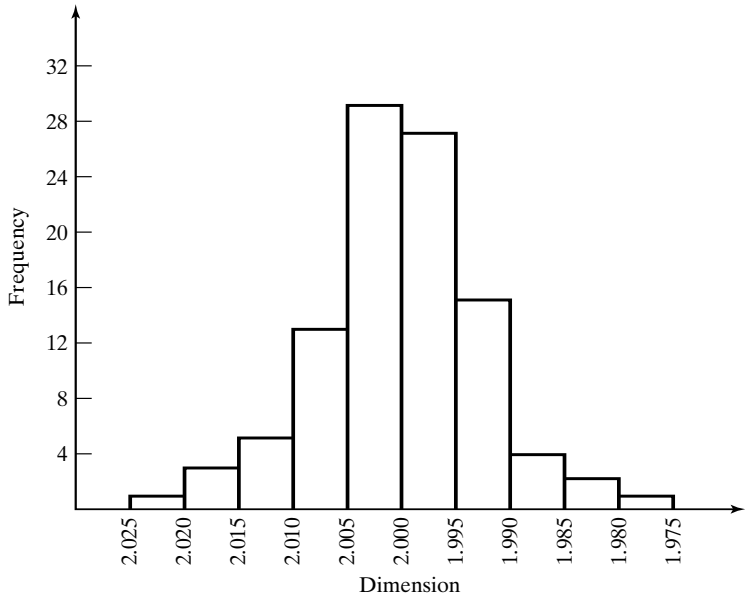


Figure 1 Histogram of the data in Table 2.

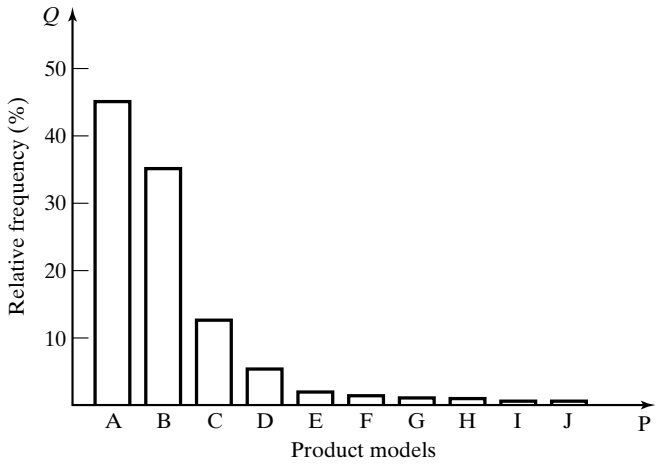


Figure 2 Typical (hypothetical) Pareto distribution of a factory's production output. Although there are ten models produced, two of the models account for 80% of the total units. This chart is sometimes referred to as a *P-Q* chart, where P = products and Q = quantity of production.

"the vital few and the trivial many."[10] The "law" was identified by Vilfredo Pareto (1848–1923), an Italian economist and sociologist who studied the distribution of wealth in Italy and found that most of it was held by a small percentage of the population.

Pareto's law applies not only to the distribution of wealth, but to many other distributions as well. The law is often identified as the 80%–20% rule (although exact percentages may differ from 80 and 20): 80% of the wealth of a nation is in the hands of 20% of its people; 80% of inventory value is accounted for by 20% of the items in inventory; 80% of sales revenues are generated by 20% of the customers; and 80% of a factory's production output is concentrated in only 20% of its product models (as in Figure 2). A Pareto chart identifies the proportion of the population that is the most important, and the focus in any improvement study or project should be on that proportion.

A Pareto distribution can also be plotted as a cumulative frequency distribution, as shown in Figure 3 for the same data shown in Figure 2. The Pareto cumulative distribution can be modeled by the following equation:[11]

$$y=\frac{(1+A)\,x}{A+x} \qquad \text{for} \qquad 0 \le y \le 1 \qquad \text{and} \qquad 0 \le x \le 1 \tag{1}$$

[10]The statement is attributed to J. M Juran [8].
[11]Based on Bender, P., "Mathematical Modeling of the 20/80 Rule: Theory and Practice," *Journal of Business Logistics*, Vol. 2, No. 2, 1981, pp 139–157.

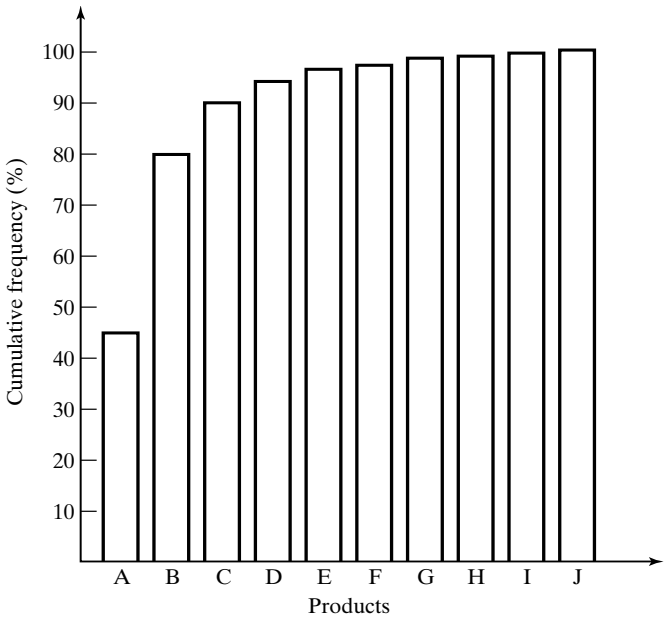


Figure 3 Pareto chart plotted as a cumulative frequency distribution for the same data shown in Figure 2.

where y = cumulative fraction of the value variable (e.g., wealth, inventory value, sales revenue), x = cumulative fraction of the item variable (e.g., population, inventory items, customers), and A is a constant that determines the shape of the distribution. Values of A between zero and infinity provide shapes that possess the Pareto characteristic, as shown in Figure 4. When $A = 0$, the equation reduces to $y = 1$ for all x, and when $A = \infty$, the equation becomes $y = x$.

To determine the appropriate value of A for a given situation or set of data, equation (1) can be rearranged to solve for A as a function of x and y, as follows:

$$A = \frac{x(1-y)}{y-x} \tag{2}$$

where x and y are the cumulative frequencies of the two variables at a given point in the distribution. The following example illustrates the approach.

Example 3 Pareto Cumulative Distribution

It is known that 20% of the total inventory items in a company's warehouse accounts for 80% of the value of the inventory. (a) Determine the parameter A in the Pareto cumulative distribution equation. (b) Given that the relationship is valid for the remaining inventory, how much of the inventory value is accounted for by 50% of the items?

Solution **(a)** To find A, we use equation (2) given that $x = 0.20$ and $y = 0.80$ (20% of the items, 80% of the value).

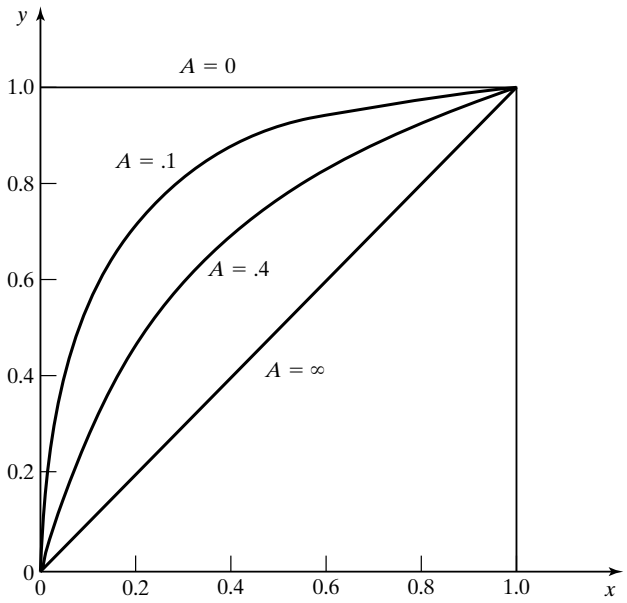


Figure 4 Shape of plots of equation (1) for several values of A.

$$A = \frac{0.20(1 - 0.80)}{0.80 - 0.20} = 0.06667$$

(b) Now that we know the value of A, the following Pareto cumulative distribution equation can be used:

$$y = \frac{(1.06667)x}{0.06667 + x}$$

For $x = 0.50$, the equation can be used to calculate y:

$$y = \frac{(1.06667)(0.50)}{0.06667 + 0.50} = 0.941$$

We expect that 50% of the items in inventory account for 94.1% of the value of the inventory. ■

Pie Charts. A ***pie chart*** is a circular (pie-shaped) display that is sliced by radii into segments whose relative areas are proportional to the magnitudes or frequencies of the data categories comprising the total circle. The visual effect is similar to a Pareto chart, in the sense that the important categories can be immediately recognized because of their relative sizes. Multiple pie charts can be displayed side by side to indicate not only the relative category sizes within a circle but also the relative sizes of the circles. Figure 5 shows two consecutive years of company sales, indicating how sales increased the second year and how the increase was distributed among types of customers.

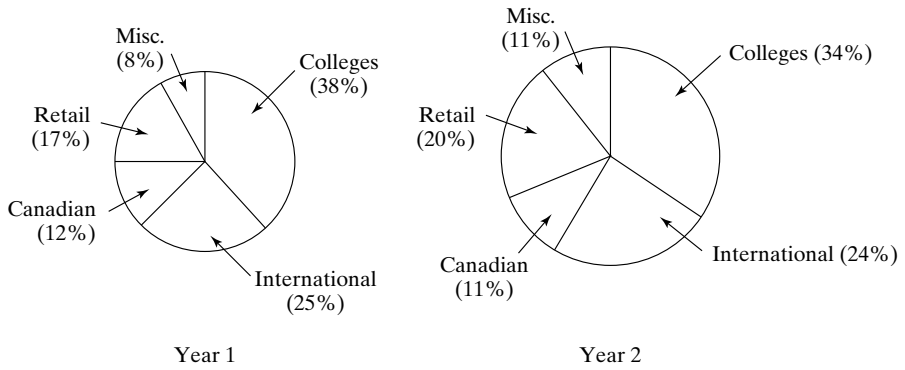


Figure 5 Pie charts showing two annual sales revenues and the customer distributions for the two years. Sales in year 2 increased 43% over year 1.

Check Sheets. The ***check sheet*** (not to be confused with "check list") is a data gathering tool generally used in the preliminary stages of the study of a problem. The operator running a process (for example, the machine operator) is often given the responsibility for recording the data on the check sheet, and the data are often recorded in the form of simple check marks.

Example 4 Check Sheet Application

For the dimensional data in the frequency distribution of Table 2, suppose we wanted to see if there were any differences between the three shifts that are responsible for making the parts. A check sheet has been designed for this purpose, and data have been collected. Analyze the data.

Solution: The check sheet is illustrated in Table 3. The data include the shift on which each dimensional value was produced (shifts are identified simply as 1, 2, and 3). The data in a check sheet are usually recorded as a function of time periods (days, weeks, months), as in our table.

It is clear from the data that the third shift is responsible for much of the variability in the data. Further analysis, shown in Table 4, substantiates this finding. This should lead to an investigation to determine the causes of the greater variability on the third shift, with appropriate corrective action to address the problem.

We also note from Table 4 that the average daily production rate for the third shift is somewhat below the daily rate for the other two shifts. The third shift seems to be a problem that demands management attention. ■

Check sheets can take many different forms, depending on the problem situation and the ingenuity of the analyst. The form should be designed to allow some interpretation of results directly from the raw data, although subsequent data analysis may be necessary to recognize trends, diagnose the problem, or identify areas of further study.

Defect Concentration Diagrams. The defect concentration diagram is a graphical method that has been found to be useful in analyzing the causes of product or part defects. It is a drawing of the product (or other item of interest), with all relevant views displayed,

TABLE 3 Check Sheet Using Data from Table 2 Recorded According to Shift on Which Parts Were Made

Range of Dimension	May 5	May 6	May 7	May 8	May 9	Weekly Totals
$1.975 \leq x < 1.980$			3			1
$1.980 \leq x < 1.985$		2		3	3	3
$1.985 \leq x < 1.990$	1	3	3	1	3	5
$1.990 \leq x < 1.995$	1 2	11 2 3	1 2	1 2	1 22	13
$1.995 \leq x < 2.000$	11 222 3	11 22 3	111 222 3	11 222 3	11 22 3	29
$2.000 \leq x < 2.005$	11 22 3	11 22 3	111 22 3	11 222 3	111 22	27
$2.005 \leq x < 2.010$	1 2 3	1 2 3	22 3	1 33	1 2 3	15
$2.010 \leq x < 2.015$	3	3	3		3	4
$2.015 \leq x < 2.020$	3			3		2
$2.020 \leq x < 2.025$	3					1
Total parts/day	20	20	21	20	19	100

TABLE 4 Summary of Data from Check Sheet of Table 3 Showing Frequency of Each Shift in Each Dimension Ranges

Range of Dimension	Shift 1	Shift 2	Shift 3	Totals
$1.975 \leq x < 1.980$			1	1
$1.980 \leq x < 1.985$		1	2	3
$1.985 \leq x < 1.990$	2		3	5
$1.990 \leq x < 1.995$	6	6	1	13
$1.995 \leq x < 2.000$	11	13	5	29
$2.000 \leq x < 2.005$	12	11	4	27
$2.005 \leq x < 2.010$	4	5	6	15
$2.010 \leq x < 2.015$			4	4
$2.015 \leq x < 2.020$			2	2
$2.020 \leq x < 2.025$			1	1
Weekly total parts/shift	35	36	29	100
Average daily parts/shift	7.0	7.2	5.8	

onto which the various types of defects or other problems of interest have been sketched at the locations where they each occurred. By analyzing the defect types and corresponding locations, it may be possible to identify the underlying causes of the defects.

Montgomery [11] describes a case study involving the final assembly of refrigerators that were plagued by surface defects. A defect concentration diagram (Figure 6) was utilized to analyze the problem. The defects were clearly shown to be concentrated around the middle section of the refrigerator. Upon investigation, it was learned that a belt was wrapped around each unit for material handling purposes. It became evident that the defects were caused by the belt, and corrective action was taken to improve the handling method.

Scatter Diagrams. In many industrial problems involving manufacturing operations, it is useful to identify a possible relationship that exists between two process variables. The scatter diagram is helpful in this regard. A ***scatter diagram*** is an *x-y* plot of the data taken of the two variables of interest, as illustrated in Figure 7. The data

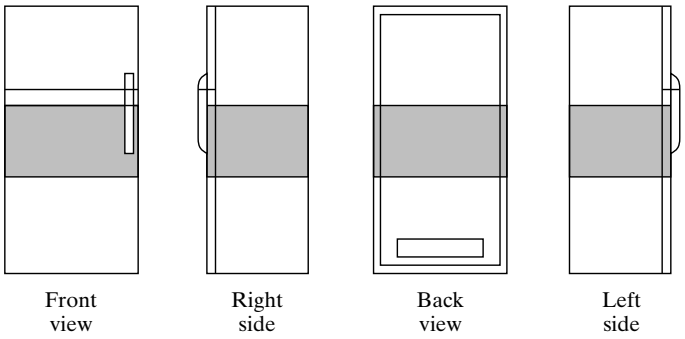


Figure 6 Defect concentration diagram showing four views of refrigerator with locations of surface defects indicated in shaded areas.

are plotted as pairs; for each x_i value, there is a corresponding y_i value. The shape of the data points considered in aggregate often reveals a pattern or relationship between the two variables. For example, the scatter diagram in Figure 7 indicates that a negative correlation exists between cobalt content and wear resistance of a cemented carbide cutting tool. As cobalt content increases, wear resistance decreases.

One must be circumspect in using scatter diagrams and in extrapolating the trends that might be indicated by the data. For instance, it might be inferred from our diagram that a cemented carbide tool with zero cobalt content would possess the highest wear resistance of all. However, cobalt serves as an essential binder in the pressing and sintering process used to fabricate cemented carbide tools, and a minimum level of cobalt is necessary to hold the tungsten carbide particles together in the final product. There are other reasons why caution is recommended in the use of the scatter diagram, since

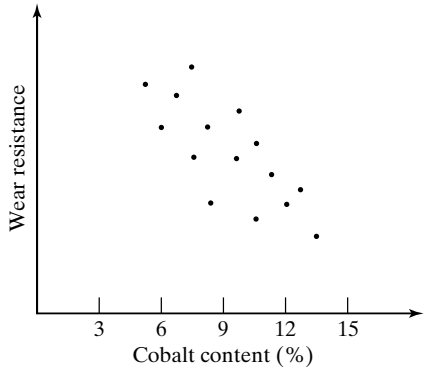


Figure 7 Scatter diagram showing the effect of cobalt binder content on wear resistance of a cemented carbide cutting tool insert.

only two variables are plotted. There may be other variables in the process whose importance in determining the output is far greater than the two variables displayed.

Cause and Effect Diagrams. The ***cause and effect diagram*** is a graphical-tabular chart used to list and analyze the potential causes of a given problem. It is not really a statistical tool, in the sense of the preceding data collection and analysis techniques. As shown in Figure 8, the diagram consists of a central stem leading to the effect (the problem), with multiple branches coming off the stem listing the various groups of possible causes of the problem. Owing to its characteristic appearance, the cause and effect diagram is also known as a ***fishbone diagram***. In application, cause and effect diagrams are often developed by worker teams who study operational problems. The diagram provides a graphical means for discussing and analyzing a problem and listing its possible causes in an organized and understandable way. Members of the team collectively identify the branches of the diagram (causes of the problem) and then attempt to determine which causes are most consequential and how to take corrective action against them.

As a starting point in identifying the causes of the problem (the main branches in the fishbone diagram), six general categories of causes are often used because they are the factors that affect performance of most production and service processes. Called the 5Ms and 1P [4], they are as follows:

1. *Machines.* This refers to the equipment and tooling used in the process.
2. *Materials.* These are the starting materials in the process.

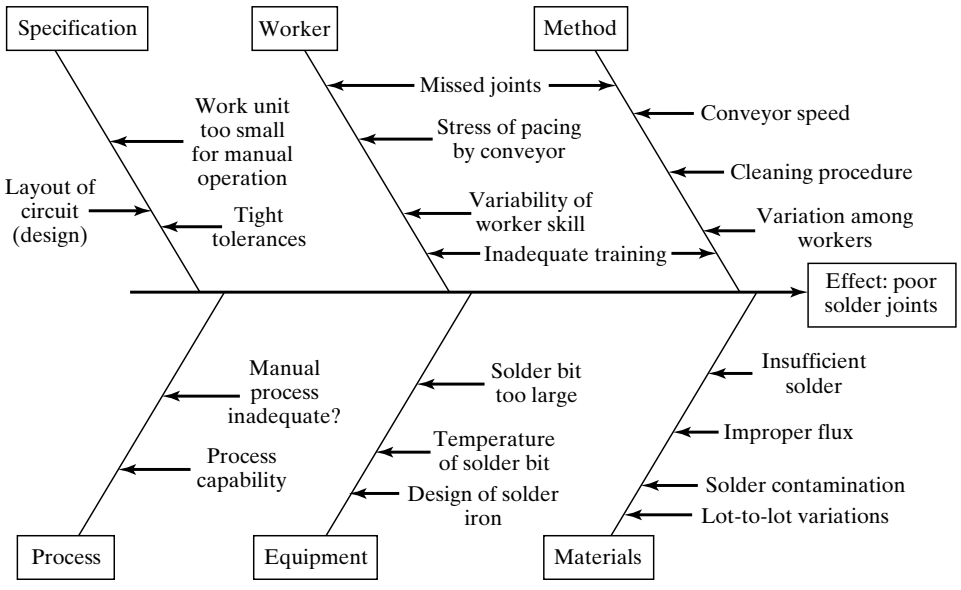


Figure 8 Cause and effect diagram for a manual soldering operation. The diagram indicates the effect (the problem is poor solder joints) at the end of the arrow and the possible causes are listed on the branches leading toward the effect.

3. *Methods.* This refers to the procedures, sequence of activities, motions, and other aspects of the method used in the process.
4. *Mother Nature.* This is a pseudonym for environmental factors such as air temperature and humidity that might affect the process.
5. *Measurement.* This relates to the validity and accuracy of the data collection procedures.
6. *People.* This is the human factor. Does the worker bring the necessary skills to the process?

During construction of the fishbone diagram, more specific causes and issues are listed on the smaller branches within each of these six categories as the analysis team pursues a solution to the problem.

4 METHODS ENGINEERING AND AUTOMATION

The issue of automation arises frequently in methods engineering. The analysis of an operation may lead to the conclusion that an automated or semiautomated work system is preferable to performing a task manually. However, a certain caution and respect must be observed in applying automation technologies. In this section, we offer three approaches for dealing with automation projects in methods engineering: (1) the USA Principle, (2) ten strategies for automation, and (3) an automation migration strategy.

4.1 USA Principle

The USA principle is a commonsense approach to automation projects. Similar procedures have been suggested in the manufacturing and automation trade literature, but none have a more captivating title than this one. USA stands for three steps in the analysis and design procedure:

1. *Understand* the existing process.
2. *Simplify* the process.
3. *Automate* the process.

Described in [10], the approach is so general that it is applicable to nearly any automation project. One might argue that the USA principle is basically an abbreviated version of the methods engineering approach (Section 2.1).

The purpose of the first step in the USA principle is to understand the current process in all of its details. What are the inputs? What are the outputs? What exactly happens to the work unit between input and output? What is the function of the process? How does it add value to the product? What are the upstream and downstream operations in the production sequence and can they be combined with the process under consideration?

Some of the basic charting tools used in methods engineering are useful in this step. Applying these kinds of tools to the existing process provides a model of the process that can be analyzed and searched for weaknesses

and opportunities. The number of steps in the process, the number and placement of inspections, the number of moves and delays experienced by the work unit, and the time spent in storage can be determined from these charting techniques.

Mathematical models of the process may also be useful to indicate relationships between input parameters and output variables. What are the important output variables? How are these output variables affected by inputs to the process, such as raw material properties, process settings, operating parameters, and environmental conditions? This information may be valuable in identifying what output variables need to be measured for feedback purposes and in formulating algorithms for automatic process control.

Once the existing process is understood, then the search begins for ways to simplify the process (step 2). This often involves a checklist of questions about the existing process. What is the purpose of this operation or this transport? Is that operation necessary? Can this step be eliminated? Is the most appropriate technology being used in this process? How can this step be simplified? Are there unnecessary steps in the process that might be eliminated without detracting from the function? These are basic questions in a methods engineering study.

Some of the ten strategies for automation (Section 4.2) may be used to simplify the process. Can steps be combined? Can steps be performed simultaneously? Can steps be integrated into a manually operated production line? Simplifying the process may lead to the conclusion that automation is not necessary, thus saving the significant investment cost that would be entailed.

When the process has been reduced to its simplest form, then automation can be considered (step 3). The possible forms of automation include those listed in the ten strategies discussed in the following section. An automation migration strategy (Section 4.3) might be implemented for a new product that has yet to prove itself.

4.2 Ten Strategies for Automation

The USA Principle is a good first step in any automation evaluation project. As suggested previously, it may turn out that automation is unnecessary or cannot be cost justified after the process has been simplified. If automation seems a feasible solution to improving productivity, quality, or another measure of performance, then the following ten strategies provide a road map to search for these improvements.[13] Although we refer to them as strategies for automation, some of them are applicable whether the process is a candidate for automation or just simplification.

1. *Specialization of operations.* Analogous to the concept of labor specialization for improving labor productivity, this strategy involves the use of special-purpose equipment designed to perform one operation with the greatest possible efficiency.
2. *Combined operations.* Production almost always occurs as a sequence of operations. Complex parts may require dozens, or even hundreds, of processing steps.

[13]These ten strategies were first published in my book *Automation, Production Systems, and Computer-Aided Manufacturing* (Prentice Hall, 1980). They seem as relevant and appropriate today as they did in 1980.

The strategy of combined operations involves reducing the number of distinct workstations through which the work units must be routed. This is accomplished by performing more than one operation at a given workstation, thereby reducing the number of separate workstations needed.

3. *Simultaneous operations.* A logical extension of the combined operations strategy is to simultaneously perform the operations that are combined at one workstation. In effect, two or more operations are performed at the same time on the same work unit, thus reducing total processing time.
4. *Integration of operations.* This strategy involves linking several workstations together into a single integrated mechanism using automated work handling devices to achieve continuous work flow. With an integrated sequence of workstations, several work units can be processed simultaneously (one at each station), thereby increasing the overall output of the system.
5. *Increased flexibility.* This strategy attempts to achieve maximum utilization of human and equipment resources for low and medium volume situations by using the same resources for a variety of work units. It involves the use of the flexible automation concepts that are implemented using computer systems.
6. *Improved material handling and storage.* The use of automated material handling and storage systems is a great opportunity to reduce nonproductive time. Typical benefits include reduced work-in-process and shorter lead times. In information service operations, the counterpart is a the use of advanced database and data processing technologies.
7. *On-line inspection.* Inspection for quality is traditionally performed after the process. This means that any poor quality product or service has already been completed by the time it is inspected. Incorporating inspection into the process permits corrections during the process. This brings the overall quality level closer to the nominal specifications intended by the designer.
8. *Process control and optimization.* This includes a wide range of control schemes intended to operate individual processes and associated equipment more efficiently. By using this strategy, the individual process times can be reduced and quality improved.
9. *Plant operations control.* Whereas the previous strategy is concerned with control of the individual process, this strategy is concerned with control at the plant level. It attempts to manage and coordinate the aggregate operations in the plant more efficiently. Its implementation usually involves a high level of computer networking within the facility.
10. *Computer integrated manufacturing* (*CIM*). Taking the previous strategy one level higher, we have the integration of factory operations with engineering design and the business functions of the firm. CIM involves extensive use of computer applications, computer databases, and computer networking throughout the enterprise.

The ten strategies constitute a checklist of the possibilities for improving the work system, through automation or simplification. They should not be considered as mutually exclusive. For many situations, multiple strategies can be implemented in one improvement project.

4.3 Automation Migration Strategy

Because of competitive pressures in the marketplace, a company often needs to introduce a new product in the shortest possible time. The easiest and least expensive way to accomplish this objective is to design a manual production method, using a sequence of workstations operating independently. The tooling for a manual method can be fabricated quickly and at low cost. If more than a single set of workstations is required to make the product in sufficient quantities, as is often the case, then the manual cell is replicated as many times as needed to meet the demand. If the product turns out to be successful and high demand is anticipated in the future, it makes sense for the company to automate production. The improvements are often carried out in phases. Many companies have an ***automation migration strategy***—a formalized plan for evolving the manufacturing systems used to produce new products as demand grows. The following phases are included in the typical automation migration strategy:

Phase 1: ***Manual production*** using single station manned cells operating independently. This is used for introduction of the new product for reasons mentioned above: quick and low-cost tooling to get started.

Phase 2: ***Automated production*** using single station automated cells operating independently. As demand for the product grows and it becomes clear that automation can be justified, the single stations are automated to reduce labor and increase production rate. Work units are still moved between workstations manually.

Phase 3: ***Automated integrated production*** using a multistation automated system with serial operations and automated transfer of work units between stations. When the company is certain that the product will be produced in mass quantities and for several years, then integration of the single station automated cells is warranted to further reduce labor and increase production rate.

This strategy is illustrated in Figure 9. Details of the automation migration strategy vary from company to company, depending on the types of products they make and the processes they perform. But well-managed companies have policies like this one. There are several advantages to such a strategy:

- It allows introduction of the new product in the shortest possible time, since production cells based on manual workstations are the easiest to design and implement.
- It allows automation to be introduced gradually (in planned phases), as demand for the product grows, engineering changes in the product are made, and time is allowed to do a thorough design job on the automated manufacturing system.
- It avoids the commitment to a high level of automation from the start, since there is always a risk that demand for the product will not justify it.

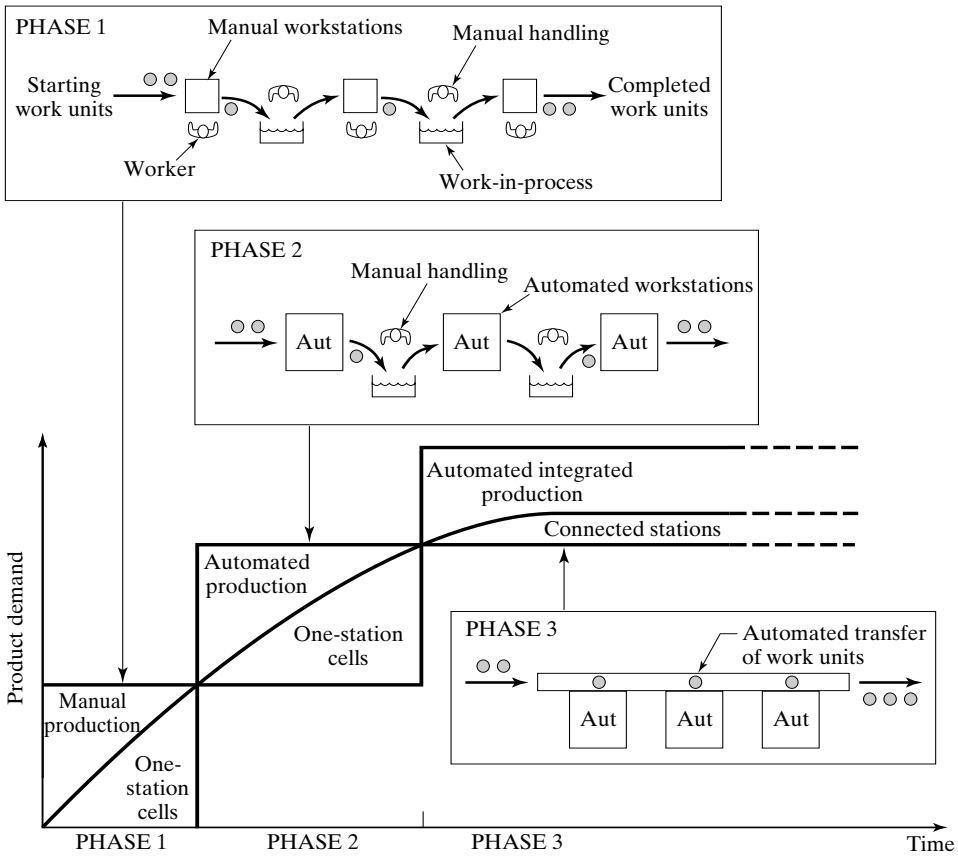


Figure 9 The three phases of a typical automation migration strategy: (1) manual production with single independent workstations, (2) automated production stations with manual handling between stations, and (3) automated integrated production with automated handling between stations.

REFERENCES

[1] Aft, L. S. *Work Measurement and Methods Improvement*. New York: Wiley, 2000.

[2] Akiyama, M., and H. Kamata. "Method Engineering and Workplace Design." Pp.4. 3–4.20 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.

[3] Barnes, R. M. *Motion and Time Study: Design and Measurement of Work*. 7th ed. New York: Wiley, 1980.

[4] Eckes, G. *Six Sigma for Everyone.* Hoboken, NJ: Wiley, 2003.

[5] Geitgey, D. C. "Operation Analysis." Pp. 3.23–3.40 in *Maynard's Industrial Engineering Handbook*, 4th ed., edited by W. K. Hodson. New York: McGraw-Hill, 1992.

[6] Groover, M., M. Weiss, R. Nagel, and N. Odrey. *Industrial Robotics: Technology, Programming,and Applications*. New York: McGraw-Hill, 1986.

[7] Groover, M. P. *Automation, Production Systems, and Computer Integrated Manufacturing*. 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001.
[8] Juran, J. M., and F. M. Gryna. *Quality Planning and Analysis*. 3rd ed. New York: McGraw-Hill, 1993.
[9] Kadota, T., and S. Sakamoto. "Methods Analysis and Design." Pp. 1415–45 in *Handbook of Industrial Engineering*, 2nd ed., edited by G. Salvendy. New York: Wiley and Institute of Industrial Engineers, Norcross, GA, 1992.
[10] Kapp, K. M. "The USA Principle." *APICS—The Performance Advantage* (June 1997): 62–66.
[11] Montgomery, D. *Introduction to Statistical Quality Control*. 3rd ed. New York: Wiley, 1996.
[12] Mundel, M. E., and D. L. Danner. *Motion and Time Study: Improving Productivity*. 7th ed. Englewood Cliffs, NJ: Prentice Hall, 1994.
[13] Muther, R. *Systematic Layout Planning*. 2nd ed. Boston, MA: Cahners Books, 1973.
[14] Niebel, B. W., and A. Freivalds. *Methods, Standards, and Work Design*. 11th ed. New York: McGraw-Hill, 2003.

REVIEW QUESTIONS

1 What is methods engineering?
2 What are the principal objectives of methods engineering?
3 What is operations analysis?
4 What was the operation studied by Frank Gilbreth in his initial research on motion study?
5 What is methods analysis?
6 What is methods design?
7 What are the six steps of the systematic approach in methods engineering?
8 The procedure offered in the text for selecting among alternatives divides the technical features of proposed equipment alternatives into two categories. What are the two categories?
9 What is a histogram?
10 What is a Pareto chart?
11 What is a check sheet?
12 What is a defect concentration diagram?
13 What is a scatter diagram?
14 What is a cause and effect diagram?
15 What does "USA" stand for in the USA principle?
16 What are the three phases in the automation migration strategy?
17 Why would a company want to use manual production methods instead of automated methods at the beginning of production of a new product?

PROBLEMS

1 A factory has 10 departments, all of which have quality problems leading to delays in shipping products to customers. A breakdown of the number of quality problems for each department (listed alphabetically) is as follows: (1) assembly, 16; (2) final packaging, 9; (3) finishing, 37; (4) forging, 73; (5) foundry, 362; (6) machine shop, 294; (7) plastic molding, 120; (8) receiving inspection, 124; (9) sheet metalworking, 86; and (10) tool-making, 42.

(a) Construct a Pareto chart for this data. (b) Assuming that all quality problems are of equal value, in which department would you start to take corrective action to reduce the quality problems? (c) Determine the percentage of total quality problems that are attributable to the two departments (20% of the departments) with the most quality problems.

2 Using your answer to part (c) of the preceding problem, (a) determine the parameter A in equation (1) representing the Pareto cumulative distribution. Use 20% of the departments as the x value in your computations. (b) Construct the idealized Pareto chart based on your answer to part (a) and discuss the comparison between this idealized chart and the actual data in the previous problem. Use a spreadsheet program to calculate the data for part (b).

3 Assume that 75% of the sales in a retail company are accounted for by 25% of the customers. (a) Determine the parameter A in the Pareto cumulative distribution equation. (b) Given that the relationship is valid for the remaining sales, how much of the sales value is accounted for by 50% of the customers?

4 The inventory policy of a retail company is to hold only the highest sales volume items in its distribution center and to ship the remaining lower sales volume items direct from the respective manufacturers to its stores. This policy is intended to reduce transportation costs. Total annual sales of the company are $1 billion. It is known that half of this amount is accounted for by only 15% of the items. In addition, it is assumed that equation (1) in the text can be used to model the Pareto cumulative distribution. (a) If the company wants to stock the top selling 35% of the items in the distribution center, what is the expected value of these items in terms of annual sales? (b) On the other hand, if the company wants to stock only those items accounting for the top 75% of annual sales, what proportion of the items corresponds to this sales volume?

5 The marketing research department for the Stitch Clothing Company has determined that 22% of the items stocked account for 70% of the dollar sales. A typical outlet store carries 1000 items. The items accounting for the top 60% of sales are replenished from the company's distribution center. The rest are shipped directly from the supplier (manufacturer) to the stores. How many items are represented by the top 60%?

6 Consider some process or procedure with which you are familiar that manifests some chronic problem. Develop a cause and effect diagram that identifies the possible causes of the problem. This is a project that lends itself to a team activity.

Charting and Diagramming Techniques for Operations Analysis

1 **Overview of Charting and Diagramming Techniques**

2 **Network Diagrams**

3 **Traditional Industrial Engineering Charting and Diagramming Techniques**
- **3.1 Operation Charts**
- **3.2 Process Charts**
- **3.3 Flow Diagrams**
- **3.4 Activity Charts**

4 **Block Diagrams and Process Maps**
- **4.1 Block Diagrams**
- **4.2 Process Maps**

Charting and diagramming techniques are useful for analyzing a work process because they graphically illustrate and summarize the activities in that process. Several charting and diagramming techniques include:

- Network diagrams for depicting work flow in sequential operations
- From-To charts for indicating material flows and/or distances among workstations or departments
- Precedence diagrams for assembly line balancing
- Gantt charts and network diagrams for scheduling projects

This chapter discusses the important charting and diagramming techniques used in methods engineering and operations analysis.

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1 OVERVIEW OF CHARTING AND DIAGRAMMING TECHNIQUES

Charting and diagramming techniques are intended to graphically display relationships among the various entities included in the graphic. Objectives in using charts and diagrams to study work include the following:

1. To permit work processes to be communicated and comprehended more readily
2. To allow the use of algorithms specifically designed for the particular diagramming technique
3. To divide a given work process into its constituent elements for analysis purposes
4. To provide a structure in the search for improvements
5. To represent a proposed new work process or method

In this chapter, the charting and diagramming techniques for operations analysis are classified into three categories: (1) network diagrams, (2) traditional industrial engineering charts and diagrams, and (3) block diagrams and process maps.

How does the analyst create the chart or diagram for the process of interest? Any of four methods or combinations thereof can be used to develop a description of the work process that is ultimately used to create the graphic:

- *The analyst is intimately familiar with the process.* In this case, the analyst works in the area and already knows how the process works. The analyst develops a graphic of the process and asks others who are also familiar with it to review it.
- *The analyst observes and records information about the process.* The analyst spends the time necessary to observe and understand the process, develops a chart or diagram of the process, and asks others familiar with it to react to it.
- *One-on-one interviews of those familiar with the process.* The analyst conducts a series of one-on-one interviews of the people who are intimately familiar with the process. A graphical depiction of the process is developed based on the interviews and subjected to review by the interviewees.
- *Group interviews of those familiar with the process.* The analyst asks those who understand the process to participate in a group meeting and describe it. A skilled facilitator may be used to elicit input from the participants. The analyst records the discussion of the meeting and develops a graphical model of the process. The analyst then asks the participants to review it. The advantage of a group meeting is that the participants can challenge each other, thus ultimately developing a more accurate picture of the actual process.

Once the chart or diagram is created, how is it analyzed? Again, there is more than one approach:

- *Algorithmic.* The specific algorithm for the particular diagram is used. Examples include line balancing algorithms for assembly lines and critical path methods for project scheduling.
- *Checklists.* In this case, the analyst reviews a series of general questions to assess whether they can be applied to the particular problem of interest. We offer a number of these checklist questions in Section 3.

- *Brainstorming.* This is a group or team activity in which participants contribute recommendations about potential improvements in the process.
- *Separating value-added and non-value-added operations.* This approach attempts to distinguish between those steps in the process that actually add value to the product or service from the customer's viewpoint and those that do not. ***Value-added steps*** are operations that (1) the customer considers important and (2) physically change the product or service. Potential non-value-added operations include rework, delays, unnecessary inspections, and unnecessary moves.

2 NETWORK DIAGRAMS

A ***network diagram*** consists of (1) nodes representing operations, work elements, or other entities and (2) arrows connecting the nodes indicating relationships among the nodes. When used in the context of work systems analysis, the arrows usually indicate either direction of work flow between nodes or precedence order among them, and the nodes represent work activities (e.g., operations, work elements, tasks).

The network diagram is used to model work flow between operations. In this case, the nodes are processing operations and the arrows indicate the sequence in which the operations are performed and the direction of work transport. A network diagram is used to model the precedence order in which work elements must be performed in assembly operations. Called a precedence diagram, nodes represent the assembly work elements, and arrows indicate the sequence in which the elements must be performed. Critical path method (CPM) and program evaluation and review technique (PERT) diagrams are network diagrams used to schedule the work activities in a project. The methods for analyzing the various network diagrams differ for the different applications, but the same basic format is used in constructing the diagrams.

For network diagrams with two-way flows between nodes (e.g., materials moving in both directions between two departments), the maximum number of arrows is given by

$$\text{Maximum number of arrows possible} = n(n - 1) \tag{1}$$

where n = number of nodes in the diagram. For network diagrams containing only one-way arrows (e.g., arrows indicating precedence order of work elements or activities), the maximum possible number of arrows between nodes in the network is given by the following:

$$\text{Maximum number of arrows possible} = \frac{n(n - 1)}{2} \tag{2}$$

Most network diagrams have fewer than the maximum values given by these equations.

3 TRADITIONAL INDUSTRIAL ENGINEERING CHARTING AND DIAGRAMMING TECHNIQUES

The charts and diagrams discussed in this section are commonly used to analyze an existing operation, sequence of operations, or other work activity for the purpose of making improvements. The underlying assumption is that by examining the work situation in detail, and subjecting the details to critical scrutiny, improvements can be found more readily than through macroscopic examination. The possible improvements include reducing cycle time and cost, eliminating unnecessary steps, mitigating safety hazards, and improving product quality. In addition to analyzing existing operations, the charts and diagrams can also be used to present proposals for new ways of accomplishing the same operations, or to design new operations that have never been implemented before.

There are many charting and diagramming techniques that have been developed over the years within the discipline of industrial engineering. For purposes of organization, we classify them into the following four major categories: (1) operation charts, (2) process charts, (3) flow diagrams, and (4) activity charts.[1] Each of these techniques provides a graphical and symbolic means of visualizing the work situation for better understanding of its scope and details. The differences relate to the levels of detail and how the work situation is represented graphically. Also, within a given category there may be variations in format and symbols depending on the subject of the analysis, for example, whether the purpose of the analysis is to study a material or to study a human worker.

In addition to these four categories of charting and diagramming techniques, the Gantt chart is a charting tool associated with traditional industrial engineering that is used in production control and project scheduling.

3.1 Operation Charts

The ***operation chart*** is a graphical and symbolic representation of the operations used to produce a product. There are two types of operations in an operation chart: (1) processing and assembly operations, and (2) inspection operations. The operation chart can also be used to analyze the steps involved in the delivery of a service, but this application of the chart is far less common. As shown in Figure 1, the operation chart consists of a series of vertical stems, each one depicting the sequence of operations and inspections performed on a given component of the product. The operation chart uses only two symbols (for operation and inspection), which are defined in Table 1. At the top of each stem is the starting material or purchased part, and the steps performed on it are indicated by symbol and brief description. The time to accomplish the operation (e.g., standard time) is also sometimes included. As each component is completed, it is assembled to other components toward the right. The column at the far right usually represents the base part or chassis of the assembly. It is the component to which all other parts in the chart are joined.

Developing the detailed listing of operations for the components and their assembly into the final entity (e.g., product, subassembly) is only the first step in the operation chart analysis. The second step involves examination of the chart to discover possible

[1]There is considerable variation in some of the terminology among different authors, and we have attempted to rationalize and abbreviate the terms. Where appropriate, we identify the alternative names of the charts.

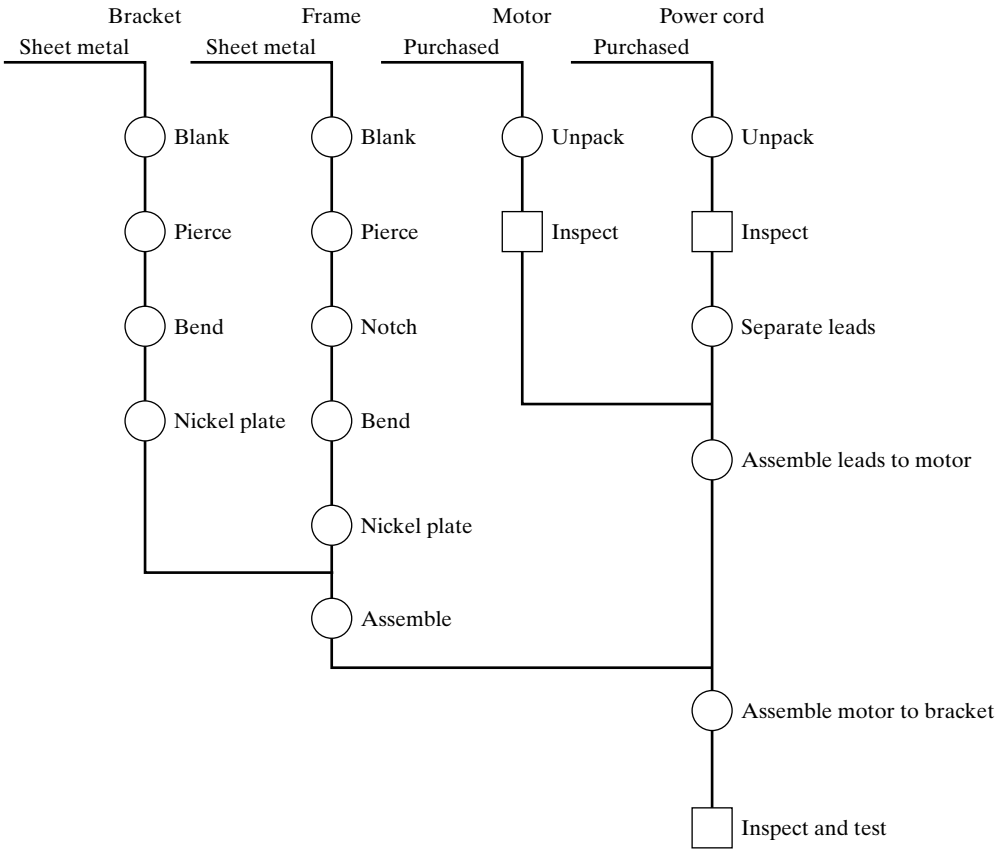


Figure 1 An operation chart for the subassembly of a product.

TABLE 1 Symbols Used in Operation Charts

Symbol	Letter	Description
○	O	***Processing or assembly operation***. Processing operations consist of changing the shape, properties, or surface of a material or workpart. Assembly operations join two or more parts to form an assembly.
□	I	***Inspection operation.*** An inspector checks the material, workpart, or assembly for quality or quantity.

improvements. Because the focus of the operation chart is on the materials of a product and the operations performed on them, the examination step consists of a questioning procedure aimed at the materials and operations. A systematic approach includes questions such as those offered in the checklist of Table 2. The third step in the use of the operation chart for an existing work situation is to develop proposals for improvement based on the results of the questioning procedure.

TABLE 2 Checklist of Questions Used to Analyze an Operation Chart

Questions related to material
What alternative starting material could be used (e.g., plastic rather than metal)?
Would a design change allow the part to be purchased as a standard commercially available item?
Could the functions of several separate components be combined into one component through a design change?
Make or buy decision: should this part be produced in our own factory or purchased from an outside vendor?
Question related to processing or assembly operations
What is the purpose of each processing operation?
Is the processing operation necessary?
Can operations be eliminated, combined, or simplified?
Is the operation time too high?
Could the processing operation be automated?
Could a different joining method be used for assembly (e.g., snap fit rather than threaded fasteners to save time)?
Questions related to inspection operations
What is the purpose of the inspection operation?
Is the inspection operation necessary?
Can the inspection operation be combined with the preceding processing or assembly operation?
If the operation is performed 100%, could it be performed on a sampling basis to reduce inspection time?
Could the inspection operation be automated?

3.2 Process Charts

A ***process chart*** is a graphical and symbolic representation of the processing activities performed on something or by somebody. The chart consists of a vertical list of the steps performed on or by the work entity using various symbols to represent operations, inspections, moves, delays, and other activities. The principal types of process chart are: (1) the ***flow process chart***, used to analyze a material or workpiece being processed, (2) the ***worker process chart***, used to analyze a worker performing a process, and (3) the ***form process chart***, used to analyze the processing of paperwork forms. These charts are described in the following sections.

Flow Process Chart. The flow process chart uses five symbols, as defined in Table 3 to detail the work performed on a material or work part as it is being processed through a sequence of operations and other activities.[2] Either iconic symbols or letter symbols can be used in constructing the chart, depending on the analyst's preference. The characteristic features of the flow process chart are shown in Figure 2. Alternative formats are possible, such as the standardized form shown in Figure 3.

The operation and inspection symbols are sometimes combined if the processing step includes a processing operation combined with an inspection at the same workstation—for example, a worker buffing a part and periodically checking its luster. In this case, the symbol consists of a circle inside a square, with the diameter of the circle equal to the side of the square.[3] Significantly more detail about the steps required to process a material

[2]Other names for the flow process chart include process chart-product analysis [7].
[3]The reverse of this format is also used in the charting literature, in which a square is positioned inside a circle to represent the combination of operation and inspection.

TABLE 3 Symbols Used in the Flow Process Chart

Symbol[a]	Letter	Description
○	O	***Operation***, usually a processing operation performed on the material at one location or workstation in which the physical shape or chemical characteristics of the material are changed. Assembly operations are unusual in a flow process chart.
□	I	***Inspection***, either to check for quality or quantity, performed at a single location or workstation.
→	M	***Move*** that involves transport of the material from one location to another, but not including moves within an operation at a workstation.
D	D	***Delay*** that occurs when the material does not or cannot proceed to the next activity—for example, a material waiting to be processed at a workstation, but other materials are ahead of it.
▽	S	***Storage*** in which the material is kept in a protected location to prevent unauthorized removal. Storage usually involves the use of a requisition to withdraw from storage, whereas a delay does not involve such a transaction.

[a]Based on symbols developed by the American Society of Mechanical Engineers (ASME). A simple arrow (→) has been substituted for the ASME move symbol (⇒) for ease of drawing.

is provided in the flow process chart than in the operation chart. Consequently, this charting technique is used to study only a single material or work part rather than the multiple components of an assembly. For each symbol, a brief description of the work activity is listed in the flow process chart. In addition, the chart also indicates the distances for move activities and times for the other activities. The time values may be especially relevant for operations, inspections, delays, and storages.

As in the operation chart, the flow process chart is examined for possible improvements and savings. However, the emphasis in the questioning procedure is expanded beyond the coverage of the operation chart because the details in the flow process chart include more types of activities. In addition to the questions shown in Table 2, current questions focus on moves, delays, and storages. Table 4 indicates the types of question that the methods analyst would want to ask. (For completeness, we have included questions from Table 2 in this checklist for the flow process chart.)

Worker Process Chart. The ***worker process chart*** is used to analyze the activities of a human worker as he or she performs a task that requires movement around a facility.[4] The symbols are virtually the same as those appearing in Table 3, but they are interpreted in terms of what the human worker does rather than what is done to a material. Table 5 summarizes the interpretations. The storage activity is difficult to interpret in the context of human work activity, so it is omitted. Analysis of the worker process chart in the search for improvements involves the same kinds of questions as in the flow process chart, only in the context of the worker performing the task of interest.

Form Process Chart. The ***form process chart*** is used to analyze the flow of paperwork forms and office procedures that normally involve the processing of documents.

[4]Alternative names for the worker process chart include process chart-person analysis [7].

Flow Process Chart					
Part No. 459011	Material: Steel C1045 forging	Description: Forgings processed in batches of 20			
Seq.	Activity Description	Symbol	Time	Distance	Analysis Notes
1	Forgings transported from forge shop	→		300 m	Forklift truck
2	Inspection of incoming forgings	□	1 hr		
3	Forgings moved and placed in storage	→		75 m	Hand truck
4	Storage	∇	7 days		Factory warehouse
5	Forgings retrieved from storage	→		75 m	Hand truck
6	Transport to machine shop	→		180 m	Forklift truck
7	Move to milling machine	→		20 m	Hand truck
8	Delay in queue for milling machine	D	5 hr		
9	Milling operation (roughing and finishing)	○	8 min/pc		Milling Machine No. 573
10	Move to drill press	→		20 m	Hand truck
11	Delay in queue for drill press	D	2 hr		
12	Drilling and tapping operations (6 holes)	○	3 min/pc		CNC Drill Press No. 226
13	Delay waiting for inspection	D	4 hr		
14	Inspection for machining operations	□	0.2 hr		
15	Delay waiting for transport to cleaning	D	3 hr		
16	Transport to finishing department	→		75 m	Forklift truck
17	Move to cleaning operation	→		10 m	Hand truck
18	Delay in queue for cleaning operation	D	30 min		
19	Cleaning operation (all parts in batch)	○	10 min		Solvent clean tank
20	Move to nickel plate operation	→		15 m	Hand truck
21	Delay in queue for nickel plate operation	D	45 min		
22	Nickel plate operation (all parts in batch)	○	20 min		Electroplating tank
23	Delay waiting for transport to storage	D	30 min		
24	Transport to storage	→		200 m	Forklift truck
25	Storage awaiting assembly	∇			Factory warehouse

Figure 2 A flow process chart used to detail the steps in the processing of a material. In the example shown, the material is a forging, and its processing consists of several machining operations, cleaning, and electroplating, but much of its time is spent in transport, delays, and storage.

Activities that occur in form processing require a change in the interpretation of the symbol. Additional symbols are also sometimes used to cover activities that are not associated with process charts for materials and workers. Symbols that can be used for the form process chart are presented and defined in Table 6.

Date:	Flow Process Chart				Page ___ of ____
Analyst:	Approval:	Summary of Activities			
Job:	Part No:	Activity (symbols)	Count	Time	Distances
Material:		Operations (○, O)			
Description:		Inspections (□ , I)			
		Moves (→, M)			
		Delays (D, D)			
		Storages (∇, S)			

Seq.	Activity Description	Symbol	Time	Distance	Analysis Notes
1					
2					
3					
4					
5					
6					

Figure 3 A standardized form for the flow process chart that can be readily created using word processing software.

3.3 Flow Diagrams

The ***flow diagram*** is a drawing of the facility layout but with the addition of lines representing movement of materials or workers to specific locations in the facility. Arrows are used on the lines to indicate the direction of movement. The flow diagram is often used in conjunction with a process chart, especially when movement of the material, worker, or form is a major factor in the analysis. An example of a flow diagram for a setup worker is presented in Figure 4. When used in connection with a process chart, the operations, inspections, delays, and storages at specific locations in the layout can be identified by numbers that are referenced to the activity numbers in the process chart.

The flow diagram reveals problems in the work flow that may not readily be identified using the process chart alone. For example, if the work flow involves considerable backtracking, this can be seen in the flow diagram whereas it is indicated only as distances in the process chart. Other work flow problems may include excessive travel, possible traffic congestion, points where delays typically occur, and inefficient layout of workstations.

3.4 Activity Charts

An ***activity chart*** is a listing of the work activities of one or more subjects (e.g., workers, machines) plotted against a time scale to indicate graphically how much time is

TABLE 4 Checklist of Questions Used to Analyze a Flow Process Chart

Questions related to material

What alternative starting material could be used (e.g., plastic rather than metal)?
Would a design change allow the part to be purchased as a standard commercially available item?
Could the functions of several separate components be combined into one component through a design change?
Make or buy decision: should this part be produced in our own factory or purchased from an outside vendor?

Question related to processing operations

What is the purpose of each processing operation?
Is the processing operation necessary?
Can operations be eliminated, combined, or simplified?
Is the operation time too high?
Could the processing operation be automated?
Where else could this operation be performed to reduce move distances?

Questions related to inspection operations

What is the purpose of the inspection operation?
Is the inspection operation necessary?
Can the inspection operation be combined with the preceding processing operation?
If the operation is performed 100%, could it be performed on a sampling basis to reduce inspection time?
Could the inspection operation be automated?

Questions relating to moves

How can moves be shortened or eliminated by combining or eliminating operations?
Can the parts or materials be collected into larger unit loads to increase efficiency in material handling?
What material handling equipment is used to transport materials? Can the level of mechanization be increased (e.g., using forklift trucks rather than manually pushed dollies)?
Can the operation sequence be modified to reduce distances traveled?

Questions relating to delays

What is the reason for the delay? Can the reason be eliminated?
Is the delay avoidable?
Why can't the material be started immediately at the next operation?

Questions relating to storage

Is the storage necessary?
Why can't the material be moved immediately to the next operation?
Can just-in-time delivery be used to eliminate the storage (e.g., can the material received from the supplier be moved immediately to the first operation after delivery rather than into storage)?

spent on each activity. The usual format is to provide brief descriptions of the activities against a vertical time scale, as shown in Figure 5 for a single worker performing a repetitive work cycle. Instead of using symbols for the work activities, as in the other charting and diagramming techniques described previously, the activities are indicated by vertical lines or bars. When bars are used, they are shaded or colored to indicate

TABLE 5 Symbols Used in the Worker Process Chart

Symbol	Letter	Description
○	O	***Operation*** performed by a worker at a single location or workstation. The operation may involve movements of materials within the workstation.
□	I	***Inspection***, either to check for quality or quantity, performed by a worker at a single location or workstation.
→	M	***Move*** in which the worker moves from one location to another as a regular element required in the task. It does not include moves within a workstation.
D	D	***Delay*** of the worker. Worker is forced by the situation to wait (e.g., waiting for an elevator). The waiting may involve moving, but the move is not a regular element required in the task (e.g., worker goes to the coffee machine while waiting for the elevator).

TABLE 6 Symbols Used in the Form Process Chart

Symbol	Letter	Description
◎	C	***Creation*** of the form (circle in a circle). This symbol is used for the origination of the form, when the form is first initiated.
○	O	***Operation*** performed on the form at a single location or workstation. The operation may involve calculations, data entries, filling out forms, folding, photocopying, stapling, assembling multiple forms into one document, etc.
□	I	***Inspection*** to read information from the form or check for correctness performed at a single location or workstation.
→	M	***Movement*** of the form from one location to another by mail or human carrier.
D	D	***Delay*** of the form. Form is waiting to be worked on, located in an in-basket or similar location other than a storage file.
▽	S	***Storage*** in a file, normally in a file cabinet or other organized filing system. This usually involves storage for a considerable time period, rather than a temporary delay.
X	X	***Disposal*** of the form. The form or a copy is destroyed.

the kind of activity being performed. Figure 6 indicates some possible shading and color conventions.[5]

Activity charts usually have more than one time scale. In Figure 5 two time scales are used, one for cumulative time during the work cycle and the second to indicate the time taken for each work activity. When activity charts are used to track several participants working together, the general name of the chart is a ***multiple-activity chart***, which consists of multiple columns, one for each participant. In this case, one time scale marks cumulative time during the cycle and a separate time scale indicates activity times for each of the columns.

There are a number of work situations in which the multiple-activity chart is useful for analysis purposes. Because the situations involve multiple entities working together, a common objective of using the chart is to analyze how the workload is coordinated and

[5]There seems to be significant variation in shading and color conventions among different experts and authors, and users are at liberty to develop their own conventions or color schemes to suit the purposes of their respective studies.

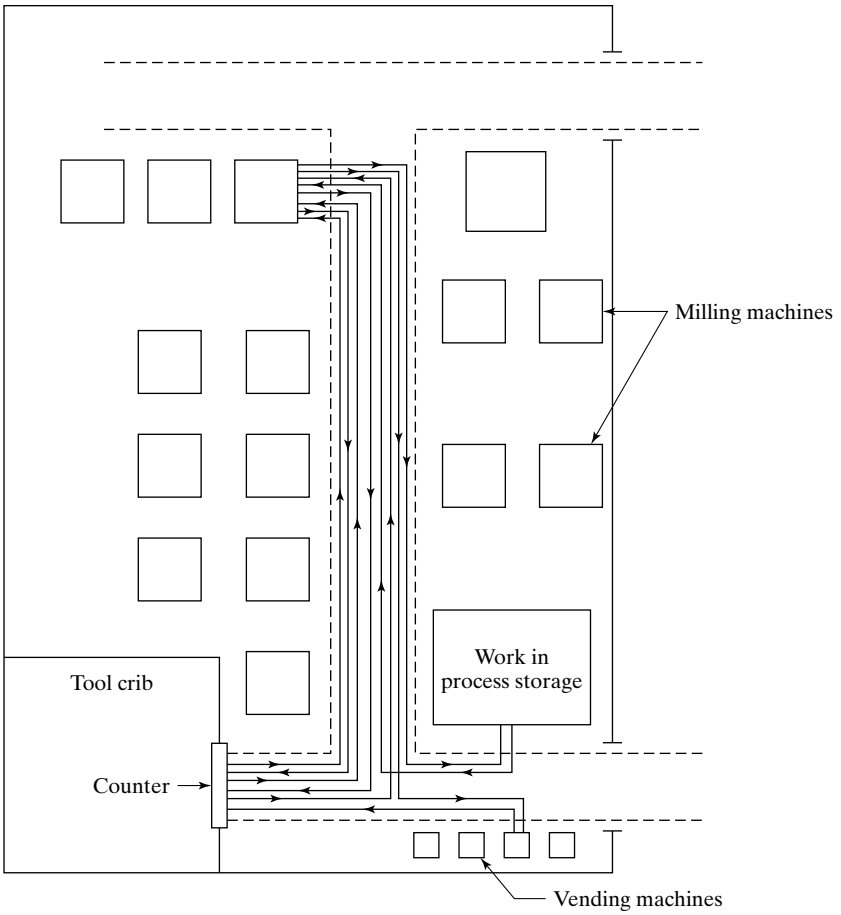


Figure 4 Flow diagram for a worker setting up a milling machine in the milling department. Note the large number of trips back and forth between the milling machine and the tool crib, which suggests that the setup worker's task might be made more efficient if all of the items needed for the setup were collected in one trip.

shared among them. The following is a listing of the major types of multiple-activity charts:

- *Right-hand/left-hand activity chart.* This chart details the contributions of the right and left hands of one worker performing a task that is highly repetitive. The task is usually performed at a single workplace, and so the chart is sometimes referred to as a ***workplace activity chart***. Figure 7 illustrates a right-hand/left-hand activity chart for a task in which the worker is using his left hand as a workholder while his right hand performs nearly all of the activities. A methods analyst would seek to install a fixture to hold the work unit during the operation

Activity Description	Chart	Activity Time (min)	Cumulative time (min)
Pick up plate from tote pan.		0.05	0.05
Carry plate to drill press and load.		0.07	0.10
Activate press.		0.03	0.15
Semiautomatic machine cycle.		0.20	0.20
			0.25
			0.30
			0.35
Remove plate.		0.03	
Carry to pallet container.		0.05	0.40
Place in pallet container.		0.02	0.45
Walk to tote pan.		0.05	0.50

Figure 5 Activity chart for one worker performing a repetitive task.

Shading	Color	Activity
Black	Blue	***Operation***: Performing an operation. Worker operating on or handling material at workplace. Machine performing an operation on automatic or mechanized cycle.
Gray	Yellow	***Inspection***: Worker performing an inspection, to check for either quantity or quality.
White (blank)	White (blank)	***Idle time***: Worker or machine is idle, waiting, or stopped.
Diagonal lines	Green	***Moving***: Worker walking outside immediate workplace (e.g., to fetch tools or materials).
Horizontal lines	Red	***Holding***: Worker holding an object in fixed position without performing any work on it.

Figure 6 Shading formats for activity charts.

and to achieve a more even balance of the workload between the right and left hands.

- *Worker-machine activity chart.* As the name indicates, this chart shows how the work elements and associated times are allocated between a worker and a machine for the repetitive cycle of a worker-machine system. Like the right-hand/left-hand activity chart, it consists of two main columns, one for the worker and the other for the machine. The worker-machine activity chart can often help to identify opportunities for cycle time improvement, such as the replacement of external work elements by internal elements, where the worker and machine perform parallel rather than sequential activities.

Left Hand	Time (min)	Right Hand	Cumulative time (min)
Pick up board	0.08		0.08
Hold board	0.06	Pick up peg and insert	0.14
Hold board	0.06	Pick up peg and insert	0.20
Hold board	0.06	Pick up peg and insert	0.26
Hold board	0.06	Pick up peg and insert	0.32
Hold board	0.06	Pick up peg and insert	0.38
Hold board	0.06	Pick up peg and insert	0.44
Hold board	0.06	Pick up peg and insert	0.50
Hold board	0.06	Pick up peg and insert	0.56
Put assembly in tote pan	0.06		0.62

Figure 7 Right-hand/left-hand activity chart.

Worker	Time (min)	Machine 1	Time	Machine 2	Time	Cumulative time (min)
Walks to machine 1	0.2					
Services machine 1	0.3	Idle	0.3			0.5
Walks to machine 2	0.2	Automatic cycle				
Services machine 2	0.3			Idle	0.3	1.0
				Automatic cycle		
	0.5					1.5
Walks to machine 1	0.2		1.2			
Services machine 1	0.3	Idle	0.3			2.0
Walks to machine 2	0.2	Automatic cycle			1.2	
Services machine 2	0.3			Idle	0.3	2.5
Idle				Automatic cycle		
	0.5					3.0
Walks to machine 1	0.2		1.2			
Services machine 1	0.3	Idle	0.3			3.5
Walks to machine 2	0.2	Automatic cycle			1.2	
Services machine 2	0.3			Idle	0.3	4.0

Figure 8 Worker-multimachine activity chart.

- *Worker-multimachine activity chart.* This chart is similar to the preceding except that the worker is responsible for more than one machine, and a work cycle must be developed that minimizes or eliminates ***machine interference*** (when one machine must wait for service because the worker is currently servicing another machine). Figure 8 illustrates the worker-multimachine activity chart for two machines. We have previously referred to this worker-multimachine arrangement as a ***machine cluster***.
- *Gang activity chart.* Other names include ***gang chart*** and ***multiworker activity chart***. This chart tracks the activities of two or more workers performing together as a team. Work situations involving the activities of a team or crew are often somewhat complex. The purpose of the activity chart analysis is to better coordinate the activities and balance the workload among the workers.

4 BLOCK DIAGRAMS AND PROCESS MAPS

Several additional diagramming techniques are available to show the detailed steps and work flow in a given process. They utilize labeled blocks and other objects, connected by arrows, to indicate processes and their interrelationships. In this section, we describe two types of diagramming techniques: block diagrams and process maps.

4.1 Block Diagrams

Block diagrams are commonly applied in linear control theory to portray the relationships among components of a physical system or process. A simple example is illustrated in Figure 9. In control system applications, arrows represent the flow of signals or variables between blocks, and blocks contain transfer functions that define how the input signals are mathematically transformed into output signals. A transfer function is the ratio of the output signal to the input signal.[6] For example, the variable y in Figure 9 equals the variable x_3 multiplied by the transfer function B.

In addition to blocks and arrows, two other operations are possible in block diagrams. First, two or more separate variables can be summed algebraically using a summing node, indicated by the circle enclosing a summation symbol in Figure 9. Thus, the variable x_2 is equal to x_1 minus z. Second, any given signal can be used as an input to more than one block. In Figure 9, the variable x_3 is used as an input to blocks B and C. The operation is called a takeoff point, shown in the diagram as a large dot between blocks A and B.

In linear control theory and other physical system applications, a form of algebra (called "block diagram algebra") can be used to reduce a complex block diagram to a single block containing the input/output relationship for the entire system. Thus, a block diagram is useful for developing the input/output functions of the individual components of a control system and then analyzing the diagram to determine overall system performance.

Our applications of block diagrams in this book deal with systems that tend to be more subjective and operations oriented. In general, our systems do not lend themselves to the mathematical treatment by which physical systems can be analyzed. Nevertheless, the block diagram format is a powerful means of depicting the signal flows and interrelationships

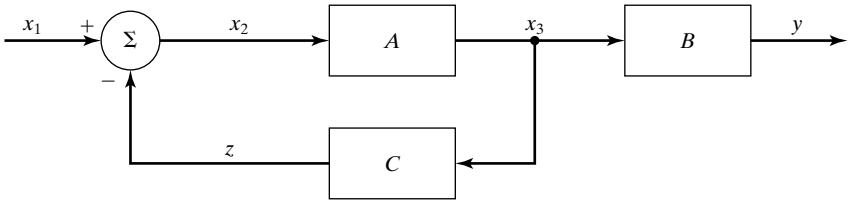


Figure 9 Block diagram of a feedback control system.

[6]To be more precise, the transfer function is the ratio of the Laplace transform of the output variable to the Laplace transform of the input variable, given that all initial conditions are zero.

among components in complex systems that are related to work. We use them frequently throughout this text.[7] In addition, block diagrams have clearly influenced the development of other diagramming techniques in operations analysis, such as process maps.

4.2 Process Maps

Process mapping techniques have been widely applied to business processes. They are also applicable to production, logistics, and service operations. As in other diagramming and charting techniques, process maps provide a detailed picture of the process or system of interest that is helpful for communicating and understanding by those involved in the operations analysis study.

There are several forms of process maps. The basic process map is a block diagram that shows the steps in the process of interest. A ***process*** is defined as a sequence of tasks or activities that add value to one or more inputs to produce outputs. The inputs and outputs may be materials, products, information, services, or other form. The process must consist of more than one step, and the steps are linked together in a logical way. The process has a beginning point and an ending point, and its purpose in the organization is to provide something of value to its customers.

Block symbols used in the basic process map are shown in Figure 10, and an example of their arrangement in a process map is illustrated in Figure 11. The block for

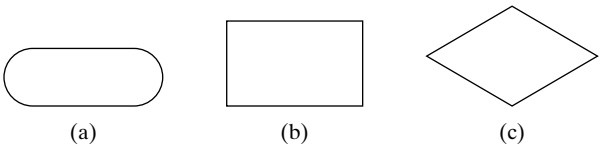


Figure 10 Symbols in the basic process map: (a) beginning/ending point of the process, (b) task or activity step, and (c) decision point. Brief labels or names are written in the various blocks to identify them.

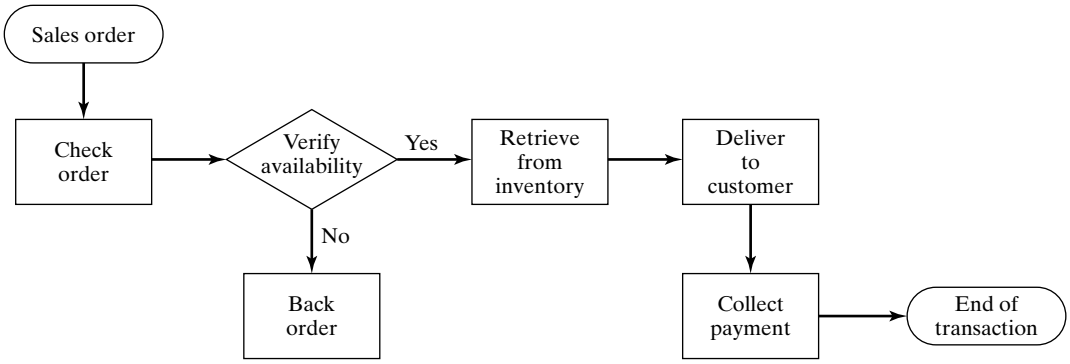


Figure 11 An example of a basic process map.

[7]The author's preference for block diagrams may derive from his educational origins in mechanical engineering.

the beginning point of the process is a racetrack oval. It is the boundary between the process under study and the activities that occur outside the process under study. The same symbol and interpretation apply to the end of the process. Rectangular blocks are used to symbolize tasks or activity steps in the process. Decision points are diamond-shaped blocks. Labels in the blocks identify what each step and decision point does, as suggested in Figure 11. Arrows are used to indicate the sequence of the steps and the flow of the work.

Process mapping can be accomplished at various levels of detail. A ***high-level process map*** depicts a macroscopic view of the process of interest and includes only the most important steps in the process. Each step in the high-level map consists of tasks or work elements, which can be examined in greater detail to create a ***low-level process map*** for that step, also called a ***detailed process map***. When this is done, the blocks in the high-level map are shadowed or otherwise highlighted to indicate that the lower-level process map exists. In some cases, more than two levels of process maps may be developed for processes requiring such detail.

Alternative forms of process maps include relationship maps and cross-functional process maps [4]. A ***relationship map*** is a block diagram that shows the input-output connections among the departments or other functional components of an organization. The map consists of blocks that represent the departments and arrows to show the flow of work. A relationship map illustrates the pairs of supplier-customer relationships throughout the organization. Every department is a customer of another department, and it is also a supplier to some other department. Arrows indicate the flow of inputs that are processed and outputs that are produced in these supplier-customer relationships. The relationship map often includes the connections with external suppliers and/or customers of the organization, as well as the associated inputs and outputs. Figure 12 illustrates a relationship map for a custom workshop.

A ***cross-functional process map*** is a block diagram that shows how the steps of a process are accomplished by the various departments or other functional groups that contribute to it. As shown in Figure 13, the departments are listed in rows separated by dashed lines. This format causes the cross-functional process map to also be known as

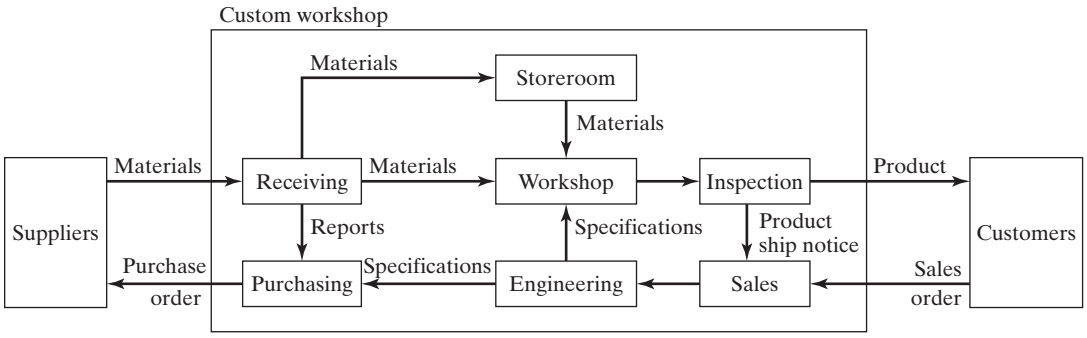


Figure 12 Relationship map for a custom workshop.

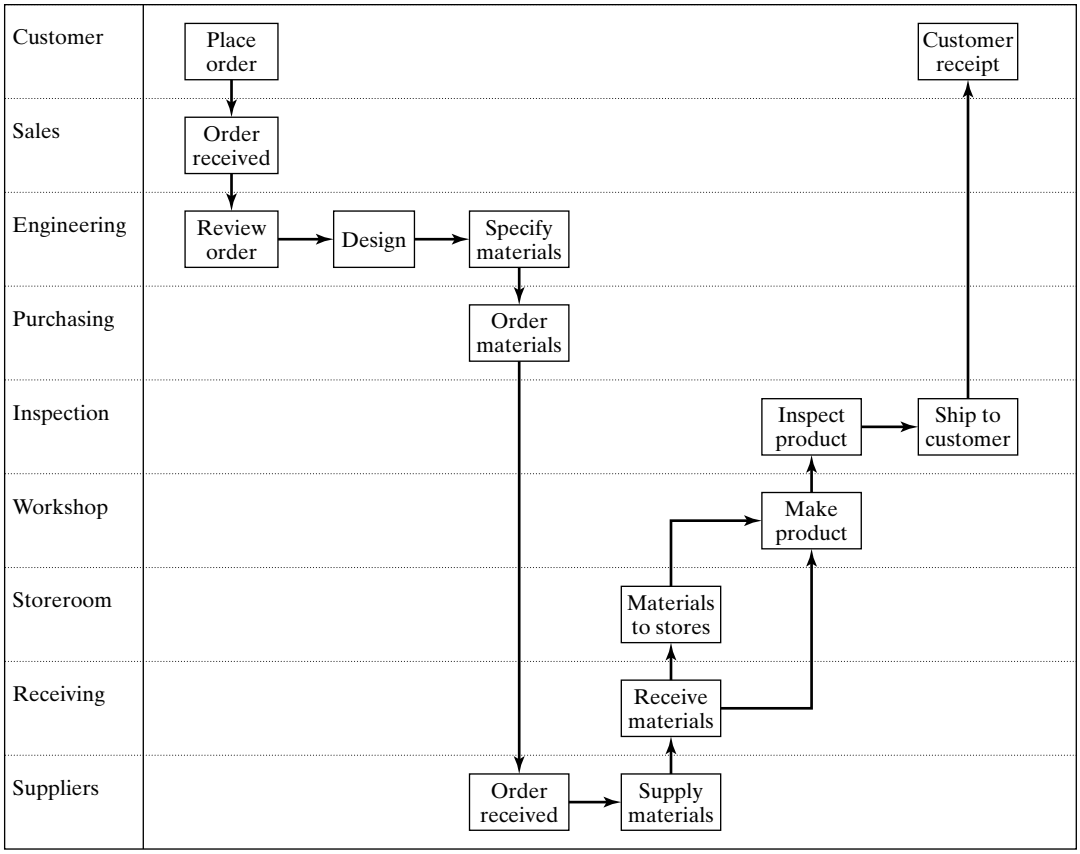


Figure 13 A cross-functional process map for a custom workshop.

the ***swim-lane chart***. Rectangular blocks represent activity steps in the process, and diamond-shaped blocks represent decision points. Arrows indicate the inputs and outputs for each block, as well as the sequence of steps.

The cross-functional process map differs from the relationship map by showing the blocks as process steps (work activities), whereas the relationship map uses blocks to represent departments. The difference between the cross-functional process map and the basic process map is the use of rows (swim lanes) to show where the process steps are accomplished. In the basic process map, the process steps do not indicate in what department the work is done.

REFERENCES

[1] Aft, L. S. *Work Measurement and Methods Improvement*. New York: Wiley, 2000.

[2] ANSI Standard Z94.0-1989. *Industrial Engineering Terminology*. Norcross, GA: Industrial Engineering and Management Press, Institute of Industrial Engineers, 1989.

[3] Barnes, R. M. *Motion and Time Study: Design and Measurement of Work*. 7th ed. New York: Wiley 1980.
[4] Damelio, R. *The Basics of Process Mapping*. Portland, OR: Productivity, Inc., 1996.
[5] Harbour, J. L. *Cycle Time Reduction: Designing and Streamlining Work for High Performance*. New York: Quality Resources, 1996.
[6] Meyers, F. E., and J. R. Stewart. *Motion and Time Study for Lean Manufacturing*. 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.
[7] Mundel, M. E., and D. L. Danner. *Motion and Time Study: Improving Productivity*. 7th ed. Englewood Cliffs, NJ: Prentice Hall, 1994.
[8] Niebel, B. W., and A. Freivalds. *Methods, Standards, and Work Design*. 11th ed. New York: McGraw-Hill, 2003.
[9] Paradiso, J. "The Essential Process." *Industrial Engineering* (April 2003): 46–48.
[10] Raymond, G. F. "Charting Procedures." Pp. 3.3–3.22 in *Maynard's Industrial Engineering Handbook*. 4th ed., edited by W. K. Hodson. New York: McGraw-Hill, 1992.
[11] Sharp, A., and P. McDermott. *Workflow Modeling: Tools for Process Improvement and Application Development*. Boston: Artech House, 2001.

REVIEW QUESTIONS

1 What are the objectives of using charts and diagrams to study work?
2 What are the four methods indicated in the text by which the analyst develops a description of the work process that is ultimately used to create the graphic?
3 What are the two characteristics of value-added steps in a given process?
4 Name some examples of network diagrams.
5 What are the two types of operations diagrammed in an operation chart?
6 Identify the five types of symbols used in a flow process chart.
7 Name the three types of process chart described in the text, and identify the application area for each.
8 What is a flow diagram?
9 What are some of the problem areas that can be identified using a flow diagram?
10 What is an activity chart?
11 Identify some of the types of multiple activity charts.
12 What are the three block symbols used in a basic process map?

PROBLEMS

Traditional Industrial Engineering Charting and Diagramming Techniques

1 A motorist experienced a flat tire on the driver side rear wheel of his car and went through the following procedure to replace the flat tire with the spare. The tire change occurred in the middle of the day in his own driveway about 20 ft in front of his garage. He first secured the other three wheels of the car with six bricks from his garage to prevent the vehicle from rolling (two trips back and forth to the garage). He then took out the jack, crank, and lug nut wrench from the trunk of the car, and read over the attached instructions for operating the jack. He removed the spare tire from the trunk and placed it near the rear left wheel.

Next, he proceeded to position the jack under the car at the recommended support beam on the car frame. He began to turn the crank to elevate the car. After the left rear portion of the car was lifted a few inches, but before the flat tire was lifted from the driveway surface, he used the lug nut wrench to loosen the five lug nuts securing the tire to the wheel hub. He then returned to the task of elevating the car, turning the jack crank until the flat tire was completely off the driveway surface. The next step was to remove the loosened lug nuts, placing them in a nearby position within reach. The flat tire was then removed and lifted into the trunk. The motorist moved the spare tire into position and lifted it onto the five studs protruding from the wheel hub. He reached for the five lug nuts, one at a time, placing them onto the studs and rotating them until finger tight. The lug nut wrench was used to tighten the five nuts. With the tire secure, he proceeded to lower the car by cranking the jack down slowly until the spare tire supported the car. For good measure, he again tightened the five lug nuts now that the car was securely on the ground. He collected the hardware, put it back into the trunk, and removed the bricks from the other three wheels and put them back into his garage. Document this tire changing procedure using a worker process chart.

2 With reference to the tire changing procedure of the previous problem, develop the steps of changing a tire, as they would be accomplished in an automotive tire center, where the worker has access to a hydraulic car lift and pneumatic lug nut wrench instead of the manual tools used by the motorist in his driveway. Document your improved tire changing procedure using a worker process chart.

3 A foundry uses the following steps in its procedure for high production of investment casting process: (1) Produce wax patterns by injection molding. (2) Transport wax patterns to an assembly work area where they are manually assembled to a wax sprue forming a pattern tree. The entire tree is made of wax. (3) Move pattern tree to a separate room where the tree is coated with a thin layer of refractory material. (4) In the same room, coat tree with successive layers of refractory material to make it a rigid structure that will become the mold for casting. (5) Move tree to a furnace room, where it should be held in an inverted position and heated to melt the wax out of the mold cavities. With the wax removed, the rigid structure is now a multiple-cavity mold with runners leading to each cavity from the sprue cavity. (6) In the same furnace room, heat the mold to a high temperature to ensure that all contaminants are removed from the mold. (7) With the mold still heated at an elevated temperature and in an upright orientation pour the molten metal into the sprue; metal will flow through the runners to each cavity. (8) After the metal cools and solidifies move the assemblage to a finishing room; the mold can be broken away from the cast metal and the parts separated from the runners and sprue. Assignment: (a) Develop the flow process chart for this casting process. (b) Based on your flow process chart, what are some changes in the investment casting procedure that you would recommend?

4 A supplier of machined components for industrial machinery (e.g., power tools, pumps, motors, compressors) operates a factory that includes a forge shop, machine shop, and finishing department. Many of the parts produced by the company are fabricated through these three departments. Because of this, the factory is laid out as three large square rooms, arranged in-line to form a rectangle with an aspect ratio of three-to-one. Each room is 200 ft by 200 ft. The rectangle runs from north to south, with the forge shop on the south end and the finishing department on the north end. Large doors are located on the south wall for work entering the factory and on the north wall for finished products exiting the factory. For one part of particular interest here, the raw material is a steel billet that is purchased from a steel wholesale supplier. The billets arrive in pallet loads of 100 billets at the shipping and receiving department, which is a building that is 35 ft by 50 ft located

25 ft from the south wall door of the factory. The shipping and receiving department inspects the parts and sends them by forklift truck to be stored in the company's warehouse that is located in another building 500 ft away from the factory in a southerly direction. The warehouse is 200 ft by 200 ft with its entrance door on the north wall. When a production order for the part is received, a factory forklift truck is dispatched to the warehouse to retrieve the billets. The forklift truck must wait while the warehouse crew locates the billets in storage, takes a pallet out of storage using the same type of forklift truck, and delivers the pallet to the dock where it is transferred to the factory forklift. The pallet is then brought back to the factory and delivered to the forge shop. The billets must wait their turn in the production schedule before being pressed into the desired shape by one of the forge presses. From the forge shop, the parts are moved to the machine shop where they are machined on two different machine tools, a milling machine and a drill press. From the machine shop, the parts travel to the finishing department for painting and baking (to cure the paint). From the finishing department, the parts are moved back to the machine shop, where additional milling is accomplished to provide two machined metal surfaces that will mate with other components in the final product. The parts are then moved to the shipping and receiving department for shipment to the customer. Develop the (a) flow process chart and (b) flow diagram for the process, using the centroid of each department to estimate distances between departments. (c) Based on your flow process chart, what are some changes in the production process that you would recommend? (d) Develop a revised procedure for the production process, documenting your revision in the form of a new flow process chart.

5 The Calm Seas Cruise Ship Line wants to analyze its passenger laundry operation. A typical cruise ship of the line is 850 ft in length and has 600 passenger cabins located on four decks. For every occupied cabin during a cruise, the cabin stewards retrieve the bed linens (sheets and pillowcases) and towels (washcloths, hand towels, and bath towels) when they make up the rooms each day. Each steward is responsible for 15 cabins. The steward separates the bed linens from the towels, putting them into two separate laundry bags. Because of the large size of each bag, the two bags are separately hand-carried to the laundry department, which is located on one of the lower decks. Depending on which deck and deck section a steward serves, the travel distance ranges from several hundred feet to more than a thousand feet, plus several flights of stairs. In the laundry department, the bags are emptied, making sure that the bed linens are not mixed with the towels during laundering. The two categories of laundry are washed separately in large washing machines, using a different wash cycle for each category. After washing, the linens and towels are dried in tumble dryers located on the same deck about 100 ft away, and then they are folded and stacked. They are then moved in stacks to the ironing department, which is located on the deck immediately above the laundry department, where they are separated and individually ironed in large flat ironers. After ironing, they are folded and stacked. From the ironing department they are transported to either of two main storerooms, located fore and aft on one of the main passenger decks for access on the following day by the cabin stewards in making up the rooms. (a) Construct a flow process chart of the laundry operation, using the bed linens and towels as the subject material in the analysis. Estimate distances moved but ignore operation times. (b) Based on your flow process chart, what are some changes in the laundry operation that you would recommend? (c) Develop a revised procedure for the laundry operation, documenting your revision in the form of a new flow process chart.

6 Given the information and data in previous problem, the cruise ship line wants to analyze the work activities of a cabin steward in its passenger laundry operation. Do not consider all of the activities that are performed in cleaning and making over a passenger cabin

(vacuuming, making beds, collecting towels and linens, etc.) as separate operations. Instead, consider making up a room and collecting the laundry in each room as one operation. The focus of the analysis is on the work activities of a cabin steward that relate to the laundry operation. (a) Construct a worker process chart to analyze the duties of the steward that are related to the ship's laundry operation. (b) Based on your worker process chart, what are some changes that you would recommend? (c) Develop a revised procedure for the laundry operation, documenting your revision in the form of a new worker process chart.

7 The Purchasing Department is required to use the following paperwork procedure for each purchase order (PO) related to a company production order. The purchase order is the legal document used by the company to order raw materials and parts from vendors in specified quantities and to guarantee payment to the vendors upon delivery of the items ordered. The purchase order procedure is triggered by the release of a production order that has been authorized by top management for one of the company's regular products. The production order indicates what product is to be produced, how many units, and when it is to be completed. To produce the product, the raw materials and component parts must be ordered in the correct quantities from suppliers (vendors). To initiate the purchase order procedure, a purchasing agent in the Purchasing Department fills out a blank purchase requisition form for each raw material or component part, obtaining the quantity information from the bill of materials for the product. The purchase requisition is an internal company document used to obtain approvals by several departments that are responsible for the product and/or its production. The purchase requisition is sent first to the Design Engineering Department where it is checked for any engineering changes that may have been made to the item ordered. The requisition is then sent from Design Engineering to the Manufacturing Engineering Department, which checks to make sure that a valid route sheet exists for the item. The route sheet is the process plan that indicates how the item is to be processed in the factory. From Manufacturing Engineering, the requisition is sent to the Production Control Department, which checks the requisition to make sure the quantity information and delivery date agrees with the production order. At each of these departments, the signature of the department manager is required in addition to the regular department employee who performed the check. It takes an average of three days in each department to obtain the necessary checks and approvals. After approval by Production Control, the purchase requisition is returned to the Purchasing Department, where it is used as the authorization to prepare the actual purchase order that will be sent to the vendor. The information on the purchase requisition is transcribed onto a blank purchase order form by the originating purchasing agent, and the PO is then signed by the manager of the Purchasing Department and mailed to the vendor. Each PO is sent in a separate envelope by first class mail, even though there are many vendors receiving more than one purchase order from the company. (a) Construct a form process chart for the current purchase order procedure. (b) Based on your form process chart, what are some changes that you would recommend? (c) Develop a revised purchase order procedure, documenting your revision in a new form process chart.

8 Consider the allocation of time between the right hand and left hand in the activity chart shown in Figure 7 in the text. (a) If the workplace were redesigned using a workholding fixture, and the worker were trained to use both hands simultaneously to perform the task, construct a right-hand/left-hand activity chart for the revised method, estimating the amounts of time for each step in the method. (b) What is the percent reduction in cycle time?

9 The repetitive work cycle in a worker-machine system consists of the work elements and associated times given in the table below. As the table shows, all of the operator's elements are external to the machine time. (a) Construct a worker-machine activity chart for this

work cycle. (b) Can some of the worker's elements be made internal to the machine cycle? If so, construct a worker-machine activity chart for the revised work cycle. What is the approximate cycle time for the revised cycle?

Sequence	Work Element Description	Worker Time (min)	Machine Time (min)
1	Worker walks to tote pan containing raw stock	0.13	(idle)
2	Worker picks up raw work part and transports it to machine	0.23	(idle)
3	Worker loads part into machine and engages machine semiautomatic cycle	0.12	(idle)
4	Machine semiautomatic cycle	(idle)	0.75
5	Worker unloads finished part from machine	0.10	(idle)
6	Worker transports finished part and deposits into tote pan	0.15	(idle)
Totals		0.73	0.75

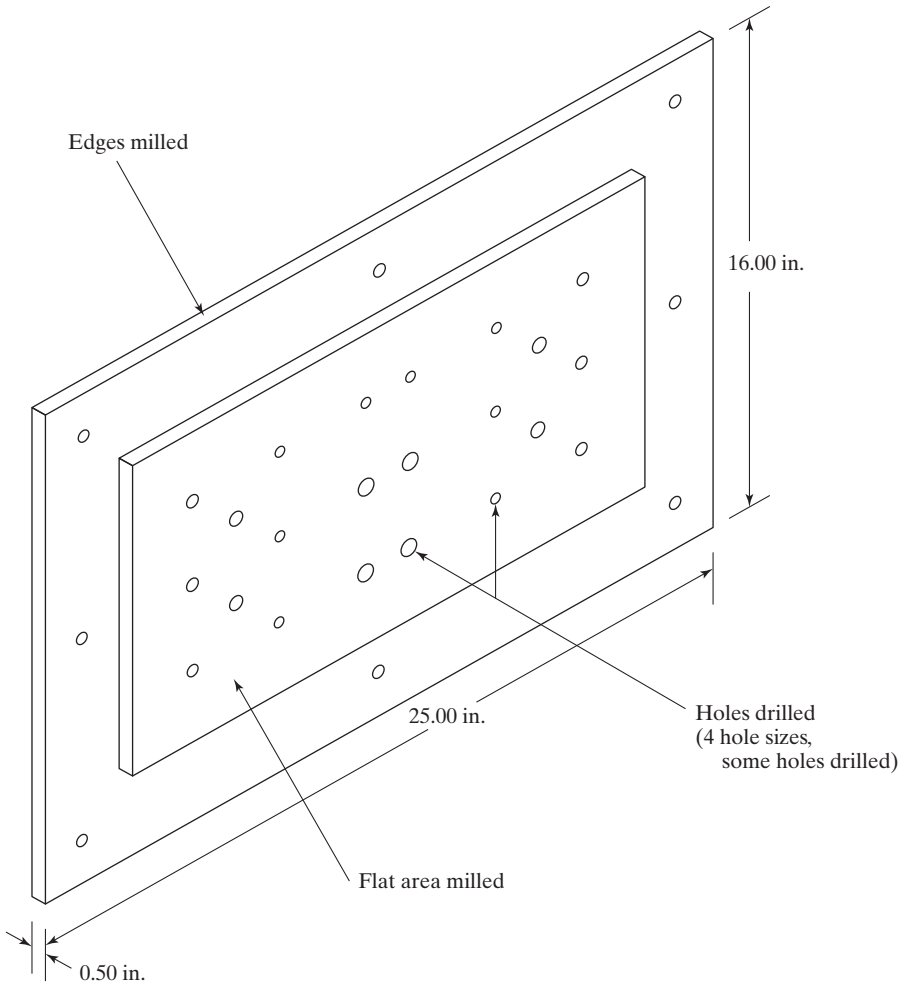
Case Problem: Flow Process Chart and Flow Diagram

10 A company produces machinery for industry. Many of the company's products are made in large quantities. This case problem deals with the mechanism plate for one of these machinery products, shown in Figure P10(a). The mechanism plate is a major part in the machine, serving as the backbone of the product. Most of the internal components and subassemblies are fastened to this plate. In the finished product, the plate is completely enclosed within the machine. The mechanism plate is the most expensive part in the product and is the source of numerous quality problems as well. Similar mechanism plates are used on many of the products, and their methods of production are similar, so any improvements made to the subject plate will apply to the other plates as well.

As is typical in batch production, the manufacturing lead time of the mechanism plate (time between release of the order and its completion) is quite long. Since other parts are being processed at the same time in the plant, there are delays in front of each production station as batches of mechanism plates wait their turn in the queues at these stations. Mechanism plates take an average of 18 weeks to complete, once a production order has been released. No information is available about how this time is distributed among the different production steps. In addition, the castings may remain in the warehouse prior to production for as much as 4 months. The following description summarizes the sequence of steps used to fabricate the mechanism plate. Figure P10(b) and (c) show floor layouts for the two stories of the factory building. Key areas and departments relating to the mechanism plate production are indicated in the description and on the floor layouts by numbers in parentheses.

The mechanism plate is a sand casting, supplied by a foundry located 20 miles away. It is made of cast iron and is very heavy, constituting fully one-third the total weight of the product. The designer selected cast iron because of its inherent rigidity and vibration damping characteristics. However, because it is a large flat part, the casting is subject to warping.

The castings are shipped to the company in batches of 50 to 100 units and inspected twice:(1) prior to being stored in the plant warehouse and (2) when an order is placed for



Part name: Mechanism Plate	Casting tol: +/−0.075 in.	Drawn by: M.P.G.
Material: Cast Iron sand casting	Machining tol: +/−0.010 in.	Date: Jan21/99

Figure p10a Exhibits for Problem 10: mechanism plate.

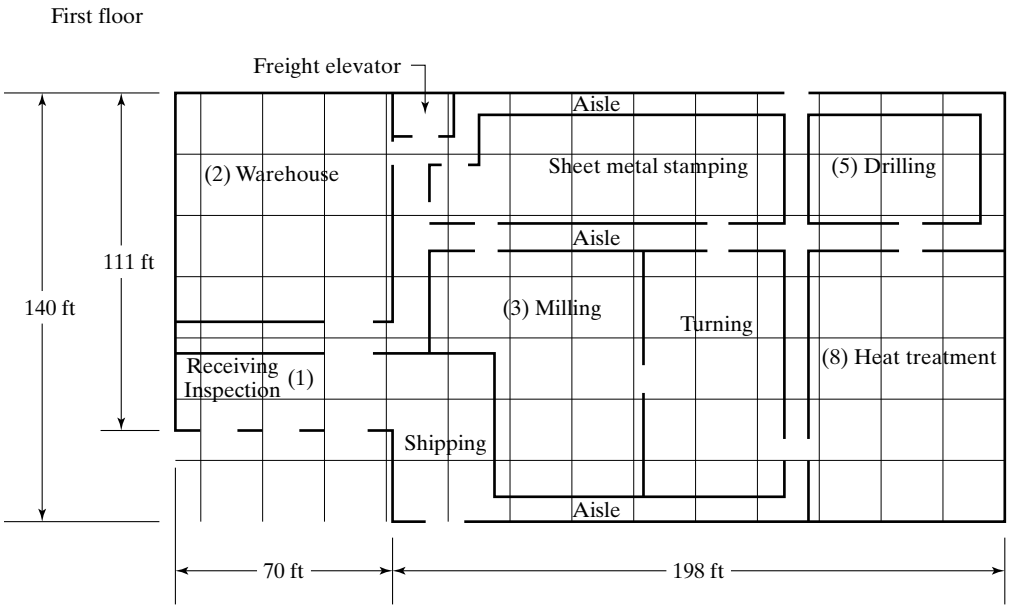


Figure p10(b) Exhibits for Problem 10: first floor layout of the factory building

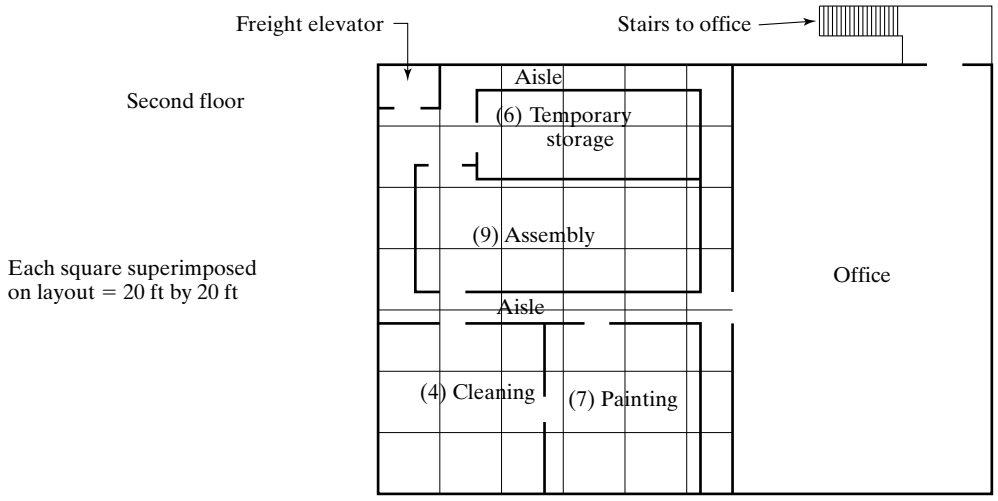


Figure p10(c) Exhibits for Problem 10: second floor layout.

their release to production. Production lot sizes are usually 5 to 10 parts. When the castings are picked from the warehouse for an order, the first step is to send them to a local firm (5 miles away) that specializes in sand blasting (which the company is not currently set up to do). This treatment dislodges any sand imbedded in the surface that is left over from the mold, removes rust that has occurred during storage, and smoothes the surface. The plates are then returned to the company warehouse.

The parts are then processed through a series of machining operations, the first and second of which are in the milling department (3). The first milling operation is face milling to make one side of the plate flat and smooth to accept the components that are to be attached. This is accomplished in one setup on a large milling machine in the mill department. The second milling requires a separate setup, and this operation makes the edges of the casting smooth and straight.

As the second milling operation is completed on each part, it is moved individually to the chemical cleaning department (4) to be cleaned and pickled. The cleaning is performed by human workers using cloth wipers soaked with chemical solvents. The purpose of the cleaning is to remove cutting fluids from machining. Pickling is then performed in open tanks filled with relatively dilute acid which etches the cast iron. The pickling brightens the surface texture of the machined surface of the casting. Owing to the size of the casting, each part must be separately dipped into the tank. The parts are then individually returned back to the machining area (drilling department) to reconstitute the batch.

The next step involves a series of hole drilling operations in the drilling department (5). These holes are used to attach the various brackets and components to the plate. This is performed on four different upright drill presses, organized according to drill size. There are four different hole sizes in the plate. After drilling, two of the hole sizes are tapped in the same department (5), using two additional drill presses. After drilling and tapping, the plates are returned to the cleaning department (4) for cleaning and pickling. Again, they must be dipped separately and moved individually to a temporary storage area (6) near the assembly department (9).

The plates are stored in the temporary storage area (6) until the other components that are to be assembled have been collected. When all of the components have been collected, the mechanism plate is sent to the painting department (7). The painting is needed to provide a protective coating and for appearance reasons. The operation is performed by human workers using paintbrushes to work the paint into the rough texture of the sand cast surface. Two coats are applied in this way, with a five hour low temperature baking operation following each coating. The baking ovens are located in the heat treatment department (8) at the rear of the factory. As many as five plates can be baked at one time.

After the final bake, the mechanism plates rejoin the other components in the temporary storage area (6). A "kit" of components is then made up, which consists of one mechanism plate and a set of all the components that are to be assembled to it. The kit is then sent to the assembly department (9), where the subassembly is completed. A total of 75 parts, including fasteners, are attached to the plate. Each mechanism plate subassembly is put together at a single workstation. When finished, it is put back into temporary storage (6) to await final assembly into the machinery product.

Assignment: (a) Prepare a flow process chart showing the operations, moves, delays, etc., in the current process. (b) Summarize the flow process chart by determining the number of operations and moves, the total distances moved, and delays, etc. (c) Construct a flow diagram of the path followed by the mechanism plate during its manufacture. (d) Develop a list of possible improvements that might be made in the production of the mechanism plate. (e) Develop a proposed improved method based on your answers to (a), (b), (c), and (d), and document your proposed new method by means of a flow process chart. (f) Summarize the flow process chart for the proposed method, indicating the number of operations and moves, the total distances traveled, and delays, etc. Show how this method is an improvement over the current method in terms of these statistics. (g) Consider the issue of plant layout in more detail. What changes in the overall layout of the plant would you make to improve the efficiency of mechanism plate production?

Process Mapping

11 Develop a basic process map for the cruise ship laundry operation described in Problem 5. What recommendations can you make for improving the operation?

12 Develop a relationship map for the machinery components factory described in Problem 4.

13 Develop a cross-functional process map for the Purchasing Department and other departments described in Problem 7. What recommendations can you make for improving the paperwork flow?

Motion Study and Work Design

1 Basic Motion Elements and Work Analysis
- **1.1 Therbligs**
- **1.2 Work Analysis Using Therbligs**

2 Principles of Motion Economy and Work Design
- **2.1 Principles Related to the Use of the Human Body**
- **2.2 Principles Related to Workplace Arrangement**
- **2.3 Principles Related to the Design of Tooling and Equipment**

This chapter focuses on the traditional industrial engineering topics of motion study and work design. ***Motion study*** involves the analysis of the basic hand, arm, and body movements of workers as they perform work.[1] ***Work design*** involves the methods and motions used to perform a task. This design includes the workplace layout and environment as well as the tooling and equipment (e.g., workholders, fixtures, hand tools, portable power tools, and machine tools). In short, work design is the design of the work system. The traditional approaches in motion study and work design are based on the findings of researchers such as Frank and Lillian Gilbreth in the early part of the twentieth century. The approaches also include common sense "principles of motion economy" that are used to simplify and improve the efficiency and effectiveness of manual work. Today, these principles are widely used not only by industrial engineers but also by workers themselves who in recent years are increasingly being given greater responsibility for designing their own work methods.

Work design is commonly associated with manual work, the type of work done in production and logistics. In recent decades, with the growth of service industries,

[1]The definition of motion study developed by the Employee and Industrial Relations Committee [ANSI Standard Z94.13-1989] is the following: "The study of the basic manual movements and procedures associated with the accomplishment of work. Historically associated with the Gilbreths (Frank and Lillian), in contrast to time study, associated with Taylor (Frederick)."

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greater attention has been focused on the design of work procedures and environments in those industries. To the extent that service operations usually involve a manual component, the general principles that apply to manual work are applicable to service and office work as well. In addition, work design principles dealing with worker comfort, workplace layout, and equipment operation apply just as well to service and office work as they do to manual tasks.

The traditional approaches in motion study and work design were developed before the advent of ergonomics and human factors, which are also applied in the design of work.

1 BASIC MOTION ELEMENTS AND WORK ANALYSIS

As indicated in the pyramidal structure of work, any manual task is composed of work elements, and the work elements can be further subdivided into basic motion elements. In this section, we define the basic motion elements and how they can be used to analyze work.

1.1 Therbligs

Frank Gilbreth was the first to catalog the basic motion elements. He called each motion a ***therblig*** (Gilbreth spelled backward except for the "th"). A list of Gilbreth's 17 therbligs is presented in Table 1 along with the letter symbol he used for each as well as a brief description. Therbligs are the basic building blocks of virtually all manual work performed at a single workplace and consisting primarily of hand motions. Therbligs are relatively few in number, but they are performed over and over, often in very similar sequences, during a given task. For example, a common hand motion sequence in repetitive assembly work is to reach for a part in the work area, grasp it, move and position it, and then release it. This sequence may be repeated many times in a work cycle, depending on how many parts are assembled in the cycle. Therbligs include mental elements as well as physical elements. With some modification, these basic motion elements are used today in a number of work measurement systems, such as Methods-Time Measurement (MTM) and the Maynard Operation Sequence Technique (MOST).

The therbligs in a manual task can be classified in several ways that are useful in analyzing the work cycle. One possible classification is shown in Table 2, in which therbligs are classified (1) as to their physical or mental emphasis and (2) whether they are effective or ineffective. Methods analysis at the therblig level seeks to eliminate or reduce ineffective therbligs.

1.2 Work Analysis Using Therbligs

Each therblig represents time and energy expended by a worker to perform a task. If the task is repetitive, of relatively short duration, and will be performed many times, it may be appropriate to analyze the therbligs that make up the work cycle as part of the

TABLE 1 Gilbreth's Seventeen Therbligs

Therblig	Letter Symbol	Description
Transport empty	TE	Reaching for an object with empty hand; for example, reach for a part prior to grasping and moving the part. Today, we commonly refer to the transport empty motion element as a "reach."
Grasp	G	Grasping an object by contacting and closing the fingers of the active hand about the object until control has been achieved.
Transport loaded	TL	Moving an object using a hand motion; for example, moving a part from one location to another at a workstation. Today, we commonly refer to a transport loaded motion element as a "move."
Hold	H	Holding an object; for example, holding an object with one hand while the other hand performs some operation on it.
Release load	RL	Releasing control of an object, typically by opening the fingers that held it and breaking contact with the object.
Preposition	PP	Positioning and/or orienting an object for the next operation and relative to an approximate location; for example, lining up a pin next to a hole for insertion into the hole. Preposition usually follows transport loaded.
Position	P	Positioning and/or orienting an object in the defined location that is intended for it. Position is generally performed during transport loaded; for example, moving a pin toward a hole and simultaneously lining it up in preparation for insertion into the hole.
Use	U	Manipulating and/or applying a tool in the intended way during the course of working, usually on an object; for example, using a screwdriver to turn a threaded fastener or using a pen to sign one's name.
Assemble	A	Joining two parts together to form an assembled entity; for example, using a threaded fastener to assemble two mating parts by hand.
Disassemble	DA	Separating multiple components that were previously joined in some way; for example, unfastening two parts held together by a threaded fastener.
Search	Sh	Attempting to find an object using the eyes or hand, concluding when the object is found.
Select	St	Choosing among several objects in a group, usually involving hand-eye coordination, and concluding when the hand has located the selected object.
Plan	Pn	Deciding on a course of action, usually consisting of short pause or hesitation in the motions of the hands and/or body.
Inspect	I	Determining the quality or characteristics of an object using the eyes and/or other senses.
Unavoidable delay	UD	Waiting due to factors beyond the control of the worker and included in the work cycle; for example, waiting for a machine to complete its feed motion.
Avoidable delay	AD	Waiting that is within the worker's control, causing idleness that is not included in the regular work cycle; for example, the worker opening a pack of chewing gum.
Rest	R	Resting to overcome fatigue, consisting of a pause in the motions of the hands and/or body during the work cycle or between cycles.

work design process. The term ***micromotion analysis*** is sometimes used for this type of analysis. For example, the activities of the right and left hands when performing a manual task can be specified in terms of therbligs. Figure 1 illustrates this kind of analysis for

TABLE 2 Classification of Therbligs

	Effective Therbligs	Ineffective Therbligs
Physical basic motion elements	Transport empty (TE) Grasp (G) Transport loaded (TL) Release load (RL) Use (U) Assemble (A) Disassemble (DA)	Hold (H) Preposition (PP)
Physical and mental basic motion elements		Position (P) Search (Sh) Select (St)
Mental basic elements	Inspect (I)	Plan (Pn)
Delay elements	Rest (R)	Unavoidable delay (UD) Avoidable delay (AD)

Source: Adapted from a classification in [7].

Sequence	Left hand	Therbligs		Right hand	Cumulative Time (min)
1	Reach for board	TE	AD	ldle	
2	Grasp board	G	AD	ldle	
3	Transport board	TL	AD	ldle	
4	Position board	PP	AD	ldle	0.08
5	Hold board	H	TE	Reach for peg	
6	Hold board	H	G	Grasp peg	
7	Hold board	H	TL	Transport peg to board	
8	Hold board	H	P	Position peg in hole	
9	Hold board	H	R	Release peg	0.14
10	Hold board	H	TE	Reach for peg	
11	Hold board	H	G	Grasp peg	
12	Hold board	H	TL	Transport peg to board	
13	Hold board	H	P	Position peg in hole	
14	Hold board	H	R	Release peg	0.20

Figure 1 Right-hand/left-hand activity chart where therbligs are used to define the work cycle. Only the first two cycles of the right hand are shown here.

the same task that is described in the right-hand/left-hand activity chart. The following general objectives are involved in micromotion analysis:[2]

- Eliminate therbligs that are ineffective if possible; for example, eliminate the need to search for parts or tools by positioning them in a known and fixed location in the workplace.

[2]The list of objectives is based largely on a list of basic principles in [11, p. 398].

- Avoid the use of a hand for holding parts; use a workholder instead.
- Combine therbligs where possible; for example, perform right-hand and left-hand motions simultaneously.
- Simplify the overall method; for example, resequence therbligs in the cycle.
- Reduce the time required for the motion; for example, shorten distances of therbligs such as transport empty and transport loaded.

As in the analysis associated with the various charting techniques, checklists are useful in micromotion analysis. Table 3 presents a checklist of questions and suggestions for possible improvements for each therblig.

2 PRINCIPLES OF MOTION ECONOMY AND WORK DESIGN

The principles of motion economy have been developed over many years of practical experience in work design. They are guidelines that can be used to help determine the work method, workplace layout, tools, and equipment that will maximize the efficiency and minimize the fatigue of the worker. Most of the guidelines are based on common sense. Many are derived from the preceding therblig analysis and will be recognizable to those who have studied Table 3. The reader might question why it is even necessary to state these principles, since some of them seem so obvious. The answers to that question can be found in factories, warehouses, offices, and other workplaces around the world in the many instances where these principles are violated.

The principles of motion economy can be organized into three categories: (1) principles related to the use of the human body, (2) principles related to workplace arrangement, and (3) principles related to the design of tooling and equipment.[3] Our coverage is based on this organization.

2.1 Principles Related to the Use of the Human Body

The principles of motion economy that are related to the use of the human body can be used to design the most appropriate methods and motions of the human worker in performing a given task. They are most applicable to manual work, either repetitive or nonrepetitive.

1. ***Both hands should be fully utilized.*** The natural tendency of most people is to use their preferred hand (right hand for right-handed people and left hand for left-handed people) to accomplish most of the work. The other hand is relegated to a minor role, such as holding the object, while the preferred hand works on it. This first principle states that both hands should be used as equally as possible.
2. ***The two hands should begin and end their motions at the same time.*** This principle follows from the first. To implement, it is sometimes necessary to design the method so that the work is evenly divided between the right-hand side and the left-hand side of the workplace. In this case, the division of work should be organized according to the following principle.

[3]This is a classification suggested by Barnes [4]. Sources for the principles discussed in this section include [1], [4], [9], [11], and [13].

TABLE 3 Micromotion Analysis Checklist for Possible Improvements

Therblig	Questions and Suggestions
Transport empty (TE)	Minimize number of parts in the product to reduce frequency of TE and TL.
	Minimize reach distance required.
	Use parts bins that have easy access.
	Can abrupt changes in direction of movement of body member be eliminated or minimized?
	Locate parts and tools used most frequently near their respective points of use.
	Minimize requirements for hand-eye coordination during reach.
	Can right and left hands be used simultaneously to accomplish two transport empty motions?
Grasp (G)	Use parts bins that have easy access.
	Use workholders that have fast release mechanism. For example, screw type vises are time consuming to operate, while pneumatic clamps are fast-acting.
	Locate parts and tools in known locations to save time in searching.
	Can right and left hands be used simultaneously to accomplish two grasp motions?
	Avoid transfer of objects from one hand to the other.
	Design parts that do not tangle.
Transport loaded (TL)	Can parts be slid across work surface rather than carried above work surface? This usually saves time.
	Can abrupt changes in direction of movement of body member be eliminated or minimized?
	Design parts and tools to be as lightweight as possible to save move time.
	Minimize number of parts in the product to reduce frequency of TE and TL.
	Minimize move distance required.
	Locate parts and tools used most frequently near their respective points of use.
	Minimize requirements for hand-eye coordination during movement.
	Can right and left hands be used simultaneously to accomplish two transport loaded motions?
Hold (H)	This is considered an ineffective therblig. Can it be eliminated?
	Can a workholding device (e.g., fixture, jig, vise, clamp) be used instead of holding by hand?
	Can friction, an adhesive, or a mechanical stop be used instead of holding by hand?
	If holding by hand cannot be eliminated, can an armrest be provided?
Release load (RL)	Is it possible to release the object by dropping it (e.g., into a chute)?
	Is the delivery point (e.g., bin, workholder) designed for ease of release of the object?
	Minimize requirements for hand-eye coordination during release.
Preposition (PP)	This is considered an ineffective therblig. Can it be eliminated?
	Can symmetry of prepositioning be increased? For example, it is easier to preposition a round shaft relative to a round hole than a square shaft relative to a square hole because of increased symmetry of the fit.
	Can a guide be designed to facilitate prepositioning?
	Can an armrest be used to steady the hand during prepositioning?
	Design parts and tools to be as lightweight as possible to save prepositioning time.
	Make sure object is grasped properly to facilitate prepositioning.
Position (P)	This is considered an ineffective therblig. Can it be eliminated?
	Can symmetry of positioning be increased? For example, it is easier to position a round shaft relative to a round hole than a square shaft relative to a square hole because of increased symmetry of the fit.
	Can a guide be designed to facilitate positioning?
	Can an armrest be used to steady the hand during positioning?
	Can tools be suspended from overhead to avoid positioning?
	Design parts and tools to be as lightweight as possible to save positioning time.
	Make sure object is grasped properly to facilitate positioning.
Use (U)	Can a more efficient hand tool be designed to reduce the time of the use motion?
	Can a portable power tool be devised to reduce the time of the use motion?

Therblig	Questions and Suggestions
	The part should be held in a workholder during the use motion.
	Can a jig be designed to guide the use of the tool? A ***jig*** is a special workholder that has a mechanism for guiding the tool.
	Can a mechanized or automated operation be used to eliminate the need for the use motion?
Assemble (A)	Can a hand tool be designed to reduce the time required for the assembly motion?
	Can a portable power tool be devised to reduce the time of the assembly motion?
	The base part or existing subassembly should be positioned in a workholder during the assembly motion.
	Can the product be designed with fewer components to minimize assembly time?
	Design the product for automated assembly to eliminate the need for manual assembly.
Disassemble (DA)	Can a hand tool be designed to reduce the time required for the disassembly motion?
	Can a portable power tool be devised to reduce the time required for the disassembly motion?
	The base part or existing subassembly should be positioned in a workholder during the disassembly motion.
Search (Sh)	This is considered an ineffective therblig. Can it be eliminated?
	Make sure lighting is adequate to facilitate searching.
	Can parts be fed from magazines or chutes to avoid searching?
	Locate tools in known positions in the workplace to facilitate searching; for example, suspend tools from overhead.
	Can different parts be made with different colors to facilitate searching?
Select (St)	This is considered an ineffective therblig. Can it be eliminated?
	Use parts bins that have easy access.
	Make sure lighting is adequate to facilitate selecting.
	Can parts be fed from magazines or chutes for one-at-a-time selection?
	Can different parts be made with different colors to facilitate searching?
Plan (Pn)	This is considered an ineffective therblig. Can it be eliminated?
	Remove the need for the worker to decide on a course of action that causes hesitation in the work cycle.
Inspect (I)	Make sure lighting is adequate to facilitate the inspection procedure.
	Minimize the number of characteristics to inspect. Only the key characteristics of the part should be inspected. Time should not be wasted inspecting unimportant characteristics.
	Can the object be inspected using gauges instead of actually measuring the characteristics of interest? Gauging takes less time than measuring.
	Can inspection be combined with another operation so it is not performed separately?
	Can inspection be automated (e.g., machine vision) to eliminate the need for a worker to accomplish it (e.g., visually)?
	Can multiple but separate inspection steps be combined into one inspection?
Unavoidable delay (UD)	This is considered an ineffective therblig. Can it be eliminated?
	Eliminate the reason for the delay. For example, can the machine speed be increased to reduce the machine cycle time?
	Can external work elements be made into internal work elements to fill up the delay time with useful work activities?
Avoidable delay (AD)	This is considered an ineffective therblig. Can it be eliminated?
	Eliminate the reason for the delay.
	Provide incentives for the worker to minimize delay time.
Rest (R)	Reduce metabolic load on worker through the use of machines and tools to minimize need for rest breaks.
	Improvements in methods and motions through analysis of previous therbligs should reduce need for rest breaks.

Source: Adapted from checklists in [4], [11], and other sources.

3. ***The motions of the hands and arms should be symmetrical and simultaneous***. This will minimize the amount of hand-eye coordination required by the worker. And since both hands are doing the same movements at the same time, less concentration will be required than if the two hands had to perform different and independent motions. It should be noted that not all tasks can be organized according to this principle. If this is the case, then the principle that follows next is applicable.
4. ***The work should be designed to emphasize the worker's preferred hand***. The preferred hand is faster, stronger, and more dexterous. If the work to be done cannot be allocated evenly between the two hands, then the method should take advantage of the worker's best hand. For example, work units should enter the workplace on the side of the worker's preferred hand and exit the workplace on the opposite side. The reason is that greater hand-eye coordination is required to initially acquire the work unit, so the worker should use the preferred hand for this element. Releasing the work unit at the end of the cycle requires less coordination.
5. ***The worker's two hands should never be idle at the same time***. The work method should be designed to avoid periods when neither hand is working. It may not be possible to completely balance the workload between the right and left hands, but it should be possible to avoid having both hands idle at the same time. The exception to this principle is during rest breaks. The work cycle of a worker-machine system may also be an exception, if the worker is responsible for monitoring the machine during its automatic cycle, and monitoring involves using the worker's cognitive senses rather than the hands. If machine monitoring is not required, then internal work elements should be assigned to the worker during the automatic cycle.

The central theme of these first five principles is the use of two hands and the allocation of work between them. Barnes [4] cites the results of a study done at the University of Iowa in which a right-handed worker performed a relatively simple manual task consisting of reaching, selecting, grasping, transporting, and releasing small parts. The task was accomplished using two different containers for the parts: a rectangular bin and a bin with tray (the bin with tray making it easier to perform the select and grasp elements). The worker performed the work cycle with each container using (1) only the right hand, (2) only the left hand, and (3) both hands performing symmetrical and simultaneous motions. The results of the study are presented in Table 4. The normalized times are simply the percentage values of the left hand and both hands using the right- hand time as the base. As expected, the right-handed worker was able to complete the work cycle in the shortest time using the right hand. Also as expected, the work cycle

TABLE 4 Study Results of a Task Performed One-Handed and Two-Handed

	Rectangular Bin			Bin with Tray		
	Right Hand	Left Hand	Both Hands	Right Hand	Left Hand	Both Hands
Time to perform	1.04 sec	1.10 sec	1.48 sec	0.81 sec	0.91 sec	1.07 sec
Normalized time	100%	106%	142%	100%	112%	132%

Source: [4].

using both hands took the longest time. However, two parts were completed using both hands, so the time per part for each tray type is one-half the value shown. Accordingly, using both hands was the most productive method.

The next five principles of motion economy attempt to utilize the laws of physics to assist in the use of the hands and arms while working.

6. ***The method should consist of smooth continuous curved motions rather than straight-line motions with abrupt changes in direction***. It takes less time to move through a sequence of smooth continuous curved paths than through a sequence of straight paths that are opposite in direction, even though the actual total distance of the curved paths may be longer (since the shortest distance between two points is a straight line). This is especially true when an object is being moved and the mass of the object affects the motions. The reason behind this principle is that the straight-line path sequence includes start and stop actions (accelerations and decelerations) that consume the worker's time and energy. Motions consisting of smooth continuous curves minimize the lost time in starts and stops.
7. ***Momentum should be used to facilitate the task wherever possible.*** When carpenters strike a nail with a hammer, they are using ***momentum***, which can be defined as mass times velocity. Imagine trying to apply a static force to press the nail into the wood. Not all work situations provide an opportunity to use momentum as a carpenter uses a hammer, but if the opportunity is present, exploit it. The previous principle dealing with smooth continuous curved motions illustrates a beneficial use of momentum (i.e., conservation of momentum) to make a task easier.
8. ***The method should take advantage of gravity instead of opposing it***. Less time and energy are required to move a heavy object from a higher elevation to a lower elevation than to move the object upward. The principle is usually implemented by proper layout and arrangement of the workplace, and so it is often associated with the workplace arrangement principles of motion economy.
9. ***The method should achieve a natural rhythm of the motions involved***. Rhythm refers to motions that have a regular recurrence and flow from one to the next. Basically, the worker learns the rhythm and performs the motions without thinking, much like the natural and instinctive motion pattern that occurs in walking.
10. ***The lowest classification of hand and arm motions should be used***. The five classifications of hand and arm motions are presented in Table 5. With each lower classification, the worker can perform the hand and arm motion more quickly and with less effort. Therefore, the work method should be composed of motions at

TABLE 5 Five Classifications of Hand and Arm Motions

Classification	Defined as:
1	Finger motions only
2	Finger and wrist motions
3	Finger, wrist, and forearm motions
4	Finger, wrist, forearm, and upper arm motions
5	Finger, wrist, forearm, upper arm, and shoulder motions

the lowest classification level possible. This can often be accomplished by locating parts and tools as close together as possible in the workplace.

The two remaining human body principles of motion economy are recommendations for using body members other than the hands and arms.

11. ***Minimize eye focus and eye travel activities***. In work situations where hand-eye coordination is required, the eyes are used to direct the actions of the hands. ***Eye focus*** occurs when the eye must adjust to a change in viewing distance—for example, from 25 in. to 10 in. with little or no change in line of sight. ***Eye travel*** occurs when the eye must adjust to a line-of-sight change—for example, from one location in the workplace to another, but the distances from the eyes are the same. Since eye focus and eye travel each take time, it is desirable to minimize the need for the worker to make these adjustments as much as possible. This can be accomplished by minimizing the distances between objects (e.g., parts and tools) that are used in the workplace.
12. ***The method should be designed to utilize the worker's feet and legs when appropriate***. The legs are stronger than the arms, although the feet are not as dexterous as the hands. The work method can sometimes be designed to take advantage of the greater strength of the legs, for example, in lifting tasks.

2.2 Principles Related to Workplace Arrangement

The principles of motion economy in this section are directed at the design of the workplace—for example, the layout of the workstation and the arrangement of tools and parts in it. The first three principles deal with the immediate work area and contribute to a natural rhythm in the work cycle. The other principles cover the use of gravity and the general conditions of the workplace.

1. ***Tools and materials should be located in fixed positions within the work area***. As the saying goes, "a place for everything, and everything in its place." The worker eventually learns the fixed locations, allowing him or her to reach for the object without wasting time looking and searching.
2. ***Tools and materials should be located close to where they are used***. This helps to minimize the distances the worker must move (travel empty and travel loaded) in the workplace. In addition, any equipment controls should also be located in close proximity. This guideline usually refers to a normal and maximum working area, as shown in Figure 2 and clarified further in Table 6. It is generally desirable to keep the parts and tools used in the work method within the normal working area, as defined for each hand and both hands working together. If the method requires the worker to move beyond the maximum working area, then the worker must move more than just the arms and hands. This expends additional energy, takes more time, and ultimately contributes to greater worker fatigue.
3. ***Tools and materials should be placed in locations that are consistent with the sequence of work elements in the work cycle***. Items should be arranged in a logical pattern that matches the sequence of work elements. Those items that are

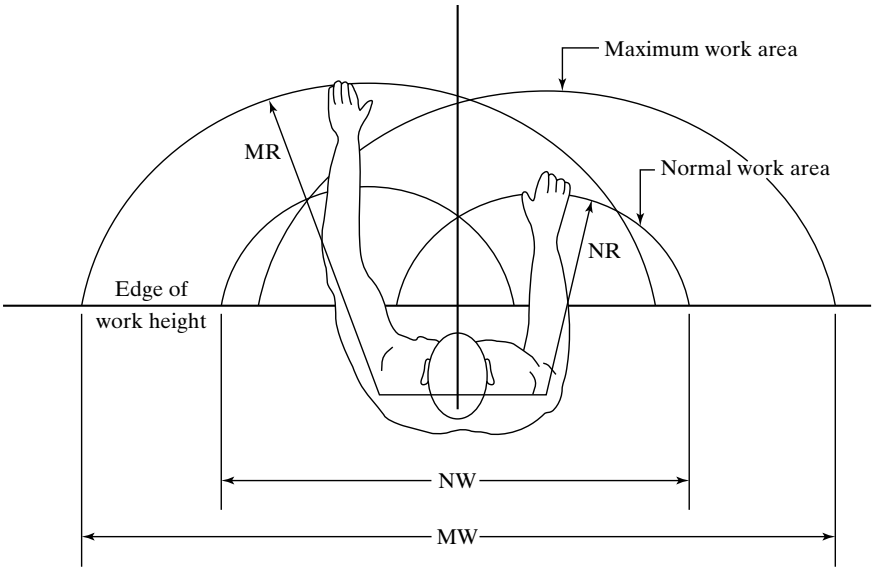


Figure 2 Normal and maximum working areas in the workplace. See Table 6 for working area dimensions of average male and female workers. (Adapted from [4]. Original source was R. Farley, "Some Principles of Methods and Motion Study as Used in Development Work," *General Motors Engineering Journal* 2, no. 6: 20–25.)

TABLE 6 Normal and Maximum Working Area Dimensions in Figure 2

Symbol in Figure 2	Dimension in Working Area for Worker Seated at Worktable	Male Worker [cm (in)]	Female Worker [cm (in)]
NR	Normal radius of arm reach	39 (15.5)	36 (14.0)
MR	Maximum radius of arm reach	67 (26.5)	60 (23.5)
NW	Normal width of arm reach	109 (43.0)	102 (40.0)
MW	Maximum width of arm reach	163 (64.0)	147 (58.0)

used first in the cycle should be on one side of the work area, the items used next should be next to the first, and so on, until the last items are obtained on the other side of the work area. The alternative to this sequential arrangement is to locate items randomly about the work area. This increases the amount of searching required and detracts from the rhythm of the work cycle.

Figure 3 shows the top view of a workplace layout that illustrates these first three principles. Note that the layout in (b) locates bins in a more accessible pattern that is consistent with the sequence of work elements.

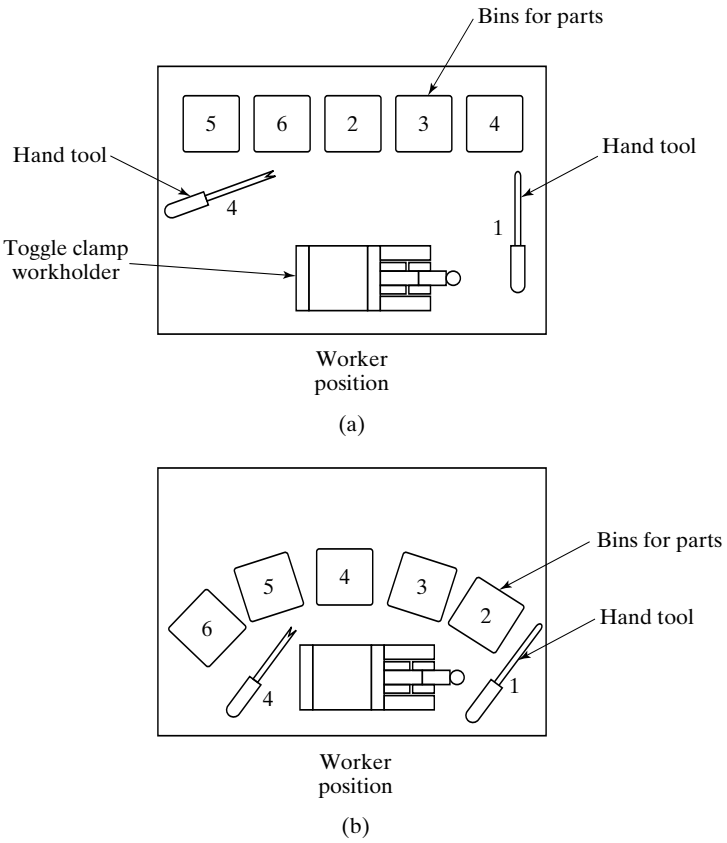


Figure 3 Two workplace layouts. The layout in (b) illustrates principles of good workplace arrangement while the one in (a) does not. Numbers indicate sequence of work elements in relation to locations of hand tools and parts bins.

4. ***Gravity feed bins should be used to deliver small parts and fasteners***. A gravity feed bin is a container that uses gravity to move the items in it to a convenient access point for the worker. One possible design is shown in Figure 4(a). It generally allows for quicker acquisition of an item than a conventional rectangular tray shown in Figure 4(b).
5. ***Gravity drop chutes should be used for completed work units where appropriate***. The drop chutes should lead to a container adjacent to the worktable. The entrance to the gravity chute should be located near the normal work area, permitting the worker to dispose of the finished work unit quickly and conveniently. There are obvious limitations on the use of gravity chutes. They are most appropriate for lightweight work units that are not fragile.

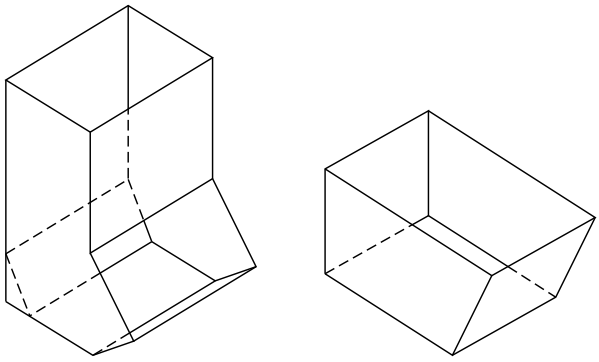


Figure 4 Two types of bins used for small parts and fasteners in the workplace: (a) gravity feed bin and (b) conventional rectangular bin.

6. ***Adequate illumination must be provided for the workplace***. The issue of illumination is normally associated with ergonomics. However, illumination has long been known to be an important factor in work design, well before ergonomics and human factors became a field of study in the scientific and engineering community. Illumination is especially important in visual inspection tasks. Adequate illumination means not only enough light on the work area; it also refers to color, absence of glare, contrast among items in the field of view, and the location of the light source(s). Other aspects of workplace environment are also important, such as noise and temperature.
7. ***A proper chair should be provided for the worker***. This usually means an adjustable chair that can be fitted to the size of the worker. The adjustments usually include seat height and back height. Both the seat and back are padded. Many adjustable chairs also provide a means of increasing and decreasing the amount of back support. The chair height should be in proper relationship with the work height. An adjustable chair for the workplace is shown in Figure 5.

2.3 Principles Related to the Design of Tooling and Equipment

The principles of motion economy in this section are helpful for designing special tools and controls on equipment used in the workplace.

1. ***Workholding devices should be designed for the task; a worker's hand should not be used as a workholder***. A mechanical workholder with a fast-acting clamp permits the work unit to be loaded quickly and frees both hands to work on the task productively. Typically, the workholder must be custom-designed for the

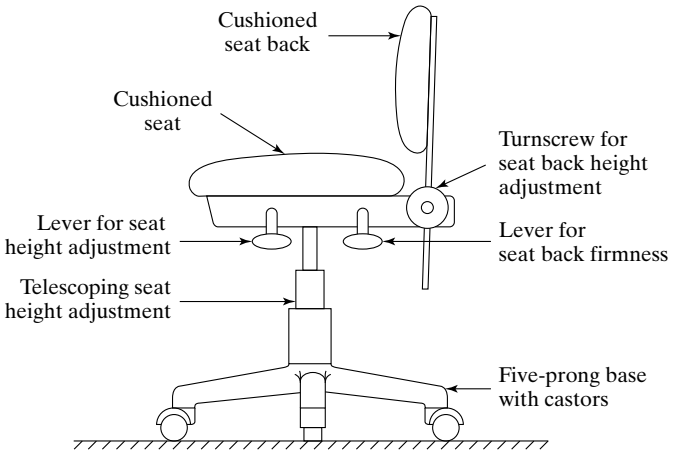


Figure 5 An adjustable chair for the workplace. The adjustments provide comfortable seating related to the size of the worker and the work height.

work part processed in the task. A conventional vise is not recommended for a repetitive work cycle because of its slow-acting screw clamping mechanism.

2. ***The hands should be relieved of work elements that can be performed by the feet***. Foot pedal controls can be provided instead of hand controls to operate certain types of equipment. Sewing machines and church organs are two examples in which foot pedals are used as integral components in the operation of the equipment. As our examples suggest, training is often required for the operator to become proficient in the use of the foot pedals.
3. ***Multiple tooling functions should be combined into one tool whenever possible***. Many of the common hand tools and implements exhibit this principle. The head of a claw hammer is designed for both striking and pulling nails. Nearly all pencils are designed for both writing and erasing. Less time is usually required to reposition such a double-function tool than to put one tool down and pick another one up.
4. ***Multiple operations should be performed simultaneously rather than sequentially whenever possible***. A work cycle is usually conceptualized as a sequence of work elements or steps. The steps are performed one after the other by the worker and/or machine. In some cases, the work method can be designed so that the steps are accomplished at the same time rather than sequentially. Special tooling and processes can often be designed to simultaneously accomplish the multiple operations. Examples include the following:

   - Pneumatically powered, multiple-spindle lug nut runner used to attach wheels to the car in automobile final assembly plants
   - Multiple-spindle drill presses to drill holes in printed circuit boards
   - Wave soldering in electronics assembly to solder all connections on a printed circuit board simultaneously.

5. ***Where feasible, an operation should be performed on multiple parts simultaneously***. This usually applies to cases involving the use of a powered tool such as a machine tool. A good example is the drilling of holes in a printed circuit board (PCB). The PCBs are stacked three or four thick, and a numerically controlled drill press drills each hole through the entire stack in one feed motion.
6. ***Equipment controls should be designed for operator convenience and error avoidance***. Equipment controls include dials, cranks, levers, switches, push buttons, and other devices that regulate the operation of the equipment. All of the controls needed by the operator should be located within easy reach, so as to minimize the body motions required to access and actuate them. In addition, the controls should be designed to be mistake-proof, so that the operator does not cause an equipment action that is different from the one intended.
7. ***Hand tools and portable powered tools should be designed for operator comfort and convenience***. For example, the tools should have handles or grips that are slightly compressible so that they can be held and used comfortably for the duration of the shift. The location of the handle or grip relative to the working end of the tool should be designed for maximum operator safety, convenience, and utility of the tool. If possible, the tool should accommodate both right-handed and left-handed workers.
8. ***Manual operations should be mechanized or automated wherever economically and technologically feasible***.[4] Mechanized or automated equipment and tooling that are designed for the specific operation will almost always outperform a human worker in terms of speed, repeatability, and accuracy. This results in higher production rates and better quality products. The economic feasibility depends on the quantities to be produced. In general, higher quantities are more likely to justify the investment in mechanization and automation. The technological feasibility relates to the time and difficulty involved in mechanizing or automating a given operation. The time issue is the amount of time required to design and build an automated machine. It may take more than a year before the equipment can be installed, but the company must begin production right away on a product with a short life cycle. The difficulty issue refers to problems such as (1) physical access to the work location, (2) adjustments required in the task, (3) requirements of manual dexterity, and (4) demands on hand-eye coordination.

REFERENCES

[1] Aft, L. S. *Work Measurement and Methods Improvement*. New York: Wiley, 2000.

[2] Akiyama, M., and H. Kamata. "Method Engineering and Workplace Design." Pp. 4.3–4.20 in *Maynard's Industrial Engineering Handbook*, 5th ed., edited by K. Zandin. New York: McGraw-Hill, 2001.

[4] ***Mechanization*** refers to the use of machinery, usually powered machinery, in place of human (or animal) labor. A human is usually required to operate and/or oversee the machinery. ***Automation*** refers to the use of powered equipment that operates under its own control and without human assistance or oversight.

[3] ANSI Standard Z94.0-1989. *Industrial Engineering Terminology*, Norcross, GA: Industrial Engineering and Management Press, Institute of Industrial Engineers, 1989.
[4] Barnes, R. M. *Motion and Time Study: Design and Measurement of Work*. 7th ed. New York: Wiley, 1980.
[5] Becker, F., and F. Steele. *Workplace by Design*. San Francisco: Jossey-Bass, 1995.
[6] Gilbreth, F. B. *Motion Study*. New York: Van Nostrand, 1911.
[7] Kadota, T., and S. Sakamoto. "Methods Analysis and Design." Pp. 1415–45 in *Handbook of Industrial Engineering*, 2nd ed., edited by G. Salvendy. New York: Wiley, and Institute of Industrial Engineers, Norcross, GA, 1992.
[8] Kanawaty, G., ed. *Introduction to Work Study*. 4th ed. Geneva, Switzerland: International Labour Office, 1992.
[9] Konz, S., and S. Johnson. *Work Design: Industrial Ergonomics*. 5th ed. Scottsdale, AZ: Holcomb Hathaway Publishers, 2000.
[10] Meyers, F. E., and J. R. Stewart. *Motion and Time Study for Lean Manufacturing*. 3rd ed. Upper Saddle River, NJ: Prentice Hall, 2002.
[11] Mundel, M. E., and D. L. Danner. *Motion and Time Study: Improving Productivity*. 7th ed. Englewood Cliffs, NJ: Prentice Hall, 1994.
[12] Nadler, G. *Work Design*. Homewood, IL: Irwin, 1963.
[13] Niebel, B. W., and A. Freivalds. *Methods, Standards, and Work Design*. 10th ed. New York: McGraw-Hill, 1999.

REVIEW QUESTIONS

1 What is motion study?
2 What is work design?
3 What is a therblig?
4 Identify some of the ineffective therbligs.
5 What is the term sometimes used for the kind of analysis involving therbligs in a task?
6 What are the general objectives of micromotion analysis?
7 Name some of the principles of motion economy that deal with the use of the two hands.
8 Why are smooth continuous curved motions better than straight-line motions when performing manual work?
9 What is the lowest classification of hand and arm motions?
10 What is the difference between eye focus and eye travel?
11 What advantage does the use of the legs have over the use of the arms?
12 What is a gravity feed bin and what is its advantage over a simple tray?
13 What are the desirable adjustments on a chair designed for the workplace?
14 Why are multiple-function tools better than separate tools for each function?

PROBLEMS

Therbligs

1 A plumber, kneeling on the floor next to the sink, reached for the monkey wrench in his toolbox, pushing around a few other tools before grasping the wrench and picking it up.

He then moved it to the pipe beneath the sink, positioned it onto the pipe fitting, and turned the fitting one-third of a rotation to loosen it. Once loose, he put down the wrench and continued turning by hand. Four rotations were required before the fitting was free of the mating pipe threads, during which he had to grasp and regrasp the fitting ten times due to the limited rotation ability of his own wrist. Write a list of the therbligs that comprise this motion sequence and label each basic motion with a brief description.

2 A secretary reached for the envelope with her right hand, picked it up and exchanged it to her left hand. She then reached for the letter opener with her right hand, picked it up, positioned the blade under the sealed lid of the envelope, and proceeded to slit the top of the envelope open. Still holding the envelope in her left hand, she laid the letter opener aside, and reached into the envelope with her right hand, pulled out the document, and began to read. Write a list of the therbligs that comprise this motion sequence and label each basic motion with a brief description.

3 A grocery store checkout clerk picked up the item from the checkout counter and moved it across the bar code scanner. Immediately, the scanner audibly responded that it had successfully identified and recorded the item. Since this was the only item the customer had brought into the checkout lane, the clerk punched one of the buttons on the cash register to print out the sales slip. While the register printed, the checkout clerk placed the item into an open plastic bag in front of him, picked up and closed the bag, and handed it to the customer. Write a list of the therbligs that comprise this motion sequence and label each basic motion with a brief description.

4 Sitting at her desk, a writer reached for a mechanical pencil, picked it up, positioned it, and then began to write on a pad of paper. After finishing one sentence, she lifted the pencil, and read the sentence. She then put the pencil aside and reached for the rectangular eraser nearby. Grasping and positioning it, she erased one of the words in the sentence. Write a list of the therbligs that comprise this motion sequence and label each basic motion with a brief description.

5 For the previous problem, identify some areas for possible study that might improve the method, indicating the nature of the improvement that might result.

6 A worker reached for a small part a short distance away in the workplace, picked up the part, and placed it in a vise. The worker then rotated the screw handle of the vise three turns to hold the part between the vise jaws. In rotating the screw handle, he had to grasp and regrasp the handle six times due to the limited rotation ability of his own wrist. On the final turn, he applied additional torque to tighten the vise. Write a list of the therbligs that comprise this motion sequence and label each basic motion with a brief description.

7 For the previous problem, identify some areas for possible study that might improve the method, indicating the nature of the improvement that might result.

8 An assembly worker reached for an Allen wrench in the workplace, hesitating momentarily while searching for the correct size from the group of Allen wrenches that were available. Finding the correct size, she picked it up and positioned it into the hexagonal socket of a screw that had previously been hand-turned into a threaded hole in the work unit. She then twirled the Allen wrench handle with one continuous finger and wrist motion until the screw had been rotated seven turns. At this point she gripped the Allen wrench handle with her hand and tightened the screw the last quarter turn. Write a list of the therbligs that comprise this motion sequence and label each basic motion with a brief description.

9 For the previous problem, identify some areas for possible study that might improve the method, indicating the nature of the improvement that might result.

Principles of Motion Economy

10 Charles Dickens was working well into the night, writing with his quill on one piece of paper after the next, the only light provided by the lone candle in front of him on the desk. (a) Identify some of the principles of motion economy that seem to be violated by Dickens' method of working and workplace. (b) Make recommendations for improving the work method and workplace.

11 An assembly worker is performing a repetitive manual task consisting of inserting 8 plastic pegs into 8 holes in a flat wooden board. A slight interference fit is involved in each insertion. Each work cycle consists of her picking up a board from the stack of boards located on the left side of the worktable (about 15 in. away from center, where they were placed by a material-handling worker), performing the 8 insertions, and then placing the assembled board into a rack that is next to the starting stack. Her first step in the work cycle is to reach for the top board in the stack with her right hand, pick it up, and exchange it to the left hand. While holding the board in her left hand, she picks up the pegs from a tray about 10 in. away in front of her with her right hand and inserts them into the holes in the board, one peg at a time. The rack holding the completed boards has a capacity of six assemblies. When the rack is full, the worker gets up from her worktable, picks up the loaded rack, carries it to a pallet located on the floor 3 ft away from the workplace, and places it onto the pallet. The pallet only holds four racks on one layer. A material-handling worker riding in a forklift truck must periodically take the pallet away and replace it with an empty pallet. Because of delays by the material-handling worker, the assembly worker is occasionally forced to stop working and wait for the pallet exchange to occur. (a) Identify some of the principles of motion economy that are violated in this work cycle. (b) What recommendations for improvement would you make?

12 Sitting in front of her all-wooden desk in her all-wooden chair, a writer reached for her wooden pencil, picked it up, positioned it, and then began to write on the pad of paper. After finishing one sentence, she lifted the pencil, and read the sentence. She then put the pencil aside and reached for the eraser nearby. Grasping and positioning it, she erased one of the words in the sentence. She then laid the eraser aside, picked the pencil up, made a correction, and continued writing. Periodically, she would repeat this sequence of writing, erasing, correcting, and then continuing to write. After working in this way for about two hours, she went and took a nap. (a) Identify some of the principles of motion economy that are violated in this work sequence. (b) What recommendations for improvement would you make?

13 The Calm Seas Cruise Ship Line wants to analyze the method used by its stewards to clean each of the passenger cabins on its ships. The current method used by a steward to clean one cabin is the following. First the steward knocks on the cabin door with his right hand to determine if the room is empty. Hearing no response, the steward reaches for the master key in his pocket, unlocks the door, opens it slightly and yells "housekeeping." If there is still no answer, he walks into the room to look around. After looking around, he goes back out into the hallway, and pushes his cart into the room. He then goes back into the hallway for his vacuum cleaner and carries it into the room. His first cleaning step is to vacuum the room and adjoining bathroom. He then strips the bed linen, puts it into the hamper on his cart, goes back into the hallway and walks down to the closet where the fresh bed linens are kept. He picks out a set of linen, carries them back to the room, and proceeds to make the bed. Next, he walks into the bathroom, picks up the used towels and carries them out to the cart hamper. He then goes back into the hallway and walks down to the same closet

as before to obtain fresh towels for the room. He brings the towels back and places them onto racks in the bathroom. From his cart, he obtains fresh soap items for the bathroom. His final cleaning step is to pick up any trash items that may have been left in the room by the passengers. (a) Identify some of the principles of motion economy that are violated in this method. (b) Besides the principles of motion economy, what other deficiencies in the method can you identify? (c) What recommendations for improvement in the method would you make?